MILITARY SPECIFICATION

CAMERA, STILL PICTURE, PANORAMIC, KA-99( )

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 5-inch (127mm) panoramic camera.

1.2 Classification. The camera shall be of one (1) type designated KA-99( ). (See 3.16).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.

MILITARY

MIL-T-27 -Transformer and Inducter (Audio, Power, and High Power Pulse), General Specification for.

MIL-P-116 -Preservation-Packaging, Methods of.
MIL-C-00675 -Coating of Glass Optical Elements (Anti-Reflection).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, ESSD (CODE 93), Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DOD-D-1000 -Drawing, Engineering and Associated List.
MIL-W-5088 -Wiring, Aircraft, Installation of.
MIL-E-5400 -Electronic Equipment, Airborne, General Specification for.
MIL-E-5558 -Enamel, Wrinkle-Finish for Aircraft Use.
MIL-M-7793 -Meter, Time Totalizing.
MIL-M-9868 -Microfilming of Engineering Documents 35mm Requirements for.
MIL-M-13508 -Mirror Front Surface Aluminized for Optical Elements.
MIL-N-18307 -Nomenclature and Identification for Electronic, Aeronautical Support Equipment Including Ground Support Equipment.
MIL-C-24308 -Connectors, Electric, Rectangular, Miniature Polarized Shell Rack and Pannel, General Specification for.
MIL-M-38510 -Microcircuits, General Specification for.
MIL-C-39006 -Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum, Established Reliability, General Specification for.

NAVAL AIR SYSTEMS COMMAND

AS-4613 -Application and Derating Requirements for Electronic Components, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 -Screw-Thread Standards for Federal Services.
FED-STD-595 -Colors.

MILITARY

MIL-STD-129 -Marking for Shipment and Storage.
MIL-STD-130 -Identification Marking of U. S. Military Property.
MIL-STD-150 -Photographic Lenses.
MIL-STD-155 -Joint Photographic Type Designation System.
MIL-STD-415 -Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-454 -Standard General Requirements for Electronic Equipment.
MIL-STD-461 -Electromagnetic Interference Characteristic Requirements for Equipment.
MIL-STD-462 -Electromagnetic Interference Characteristics Measurement of.
MIL-STD-470 -Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471 -Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-681 -Identification Coding and Application of Hookup and Lead Wire.
MIL-STD-704 -Electric Power, Aircraft, Characteristics and Utilization of.

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MIL-STD-781 -Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-785 -Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810 -Environmental Test Methods.
MIL-STD-882 -System Safety Program Requirements.
MIL-STD-883 -Test Methods and Procedures for Microelectronics.
MIL-STD-965 -Parts Control Program.
MIL-STD-1278 -Filters, Light, Photographic.
MIL-STD-1304 -Reports: Reliability and Maintainability Engineering Data.
MIL-STD-1353 -Electrical Connectors, and Associated Hardware, Selection and Use of.
MIL-STD-1390 -Level of Repair.
MIL-STD-1472 -Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1562 -Lists of Standard Microcircuits.
MIL-STD-2070 -Procedures for Performing a Failure Mode Effects and Criticality Analysis for Aeronautical Equipment.
MIL-STD-2074 -Failure Classification for Reliability Testing.
MIL-STD-2077 -Test Program Sets, General Requirements for.
MS-26565 -Spool, Photographic Film Aerial.

DRAWINGS

NAVAL AIR DEVELOPMENT CENTER

NADC 822-2046 -KA-99 Cradle Assembly.

HANDBOOKS

MILITARY

MIL-HDBK-217 -Reliability Stress and Failure Rate Data for Electronic Equipment.
MIL-HDBK-472 -Maintainability Prediction.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

NAVAIR 16-1-525 -Preferred Standard Avionics Test Equipment.
NAVAIR 19-1-127 -Preferred List of Common Ground Support Equipment (Nonavionic).
AR-5 -Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of.
AR-10 -Maintainability of Avionics Equipment and Systems, General Requirements for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARD INSTITUTE

PH1.10-1976 -Dimensions for Unperforated and Perforated Photographic Film in Rolls, Including Leaders and Trailers, for Aerial and Related Uses.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample camera(s) for first article inspection and approval (see 4.5 and 6.3).

3.2 Parts, materials and processes. The selection and application of parts, materials, and processes shall be in accordance with MIL-E-5400, except as specified herein.

3.2.1 Screening of parts. All electrical, electronic and electromechanical parts shall be subjected to 100 percent screening. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similar military standard or standard-established reliability part, as well as the requirements of Table I. All integrated circuits shall be screened to MIL-STD-883, Method 5004, Class B as a minimum. Hybrid circuits shall be screened to MIL-STD-883, Method 5008 as a minimum.

3.2.2. Parts derating and application. Application and derating requirements shall be in accordance with AS-4613.

3.2.2.1 Application limitations. Limitations on parts usage shall be as specified herein.

3.2.2.1.1 Prohibited application. The use of electron tubes, aluminum electrolytic capacitors, fixed paper and plastic dielectric capacitors in molded cases, plastic epoxy or other encapsulated semiconductors, germanium semiconductors, and photo couplers (photo transistors) shall be prohibited.

3.2.2.1.2 Restricted application The use of nonsolid sintered-slug tantalum capacitors is restricted to MIL-C-39006, CLR 79 qualified devices. Programmable Read Only Memories (PROMs) and hot carrier (Schottky) rectifiers shall not be used in production equipment.

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TABLE I. Supplemental part screening.

	100% Environmental Screen					100% Burn-in	
	Temp. Cycling	High Temp.	Case Leakage	Cent-rifuge	Other	Time Temp.	Parameters
Paper and/or plastic dielectric caps	-85°F(-65°C) 257°F(125°C) 10 cycles					100±5 hr. at rated voltage and max. rated temp.	C, Insul. resistance DF, Dielectric withstanding voltage.
Fixed metalized paper dielectric D. C.	-76°F(-60°C) 257°F(125°C) 10 cycles					100±5 hr. at rated voltage at max. ambient temp.	C, Insul. resistance DF, Dielectric withstanding voltage.
Feed thru RF det AC & DC	-85°F(-65°C) 257°F(125°C) 10 cycles					100±5 hr. at rated voltage at max. ambient temp.	C, Insul. resistance DF, Dielectric withstanding voltage.
Resistors, carbon comp. and others.	-85°F(-65°C) 257°F(125°C) 10 cycles				***		
Diodes	-85°F(-65°C) 392°F(200°C) 10 cycles	392°F(200°C)	Pressure bomb	20KG	Power Pulse (Pre-Cap Visual	160 hr. at max. rated power at 257°F(125°C)	Group A of nearest military established part.
Transistors	-85°F(-65°C) 392°F(200°C) 10 cycles	392°F(200°C)	$5X10^{-8}$ cc/sec	20KG	Rev Bias 302°F(150°C) (PNP) (Pre-Cap) Visual	168 hr. at max. rated power at 257°F(125°C) HTRB PHP only	Group A of nearest military established part.

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TABLE I. Supplemental part screening.(2) - Continued

	100% Environmental Screen				100% Burn-in		
	Temp. Cycling	High Temp.	Case Leakage	Cent-rifuge	Other	Time Temp.	Parameters
Integrated Circuits					MIL-STD-883 Method 5004, Class B MIL-STD-883 Method 5008		
Hybrids							
Transformers and Inductors	-67°F (-55°C) 257°F (125°C) 10 cycles						Group A inspection per MIL-T-27

(1) - Intended primarily for state-of-the-art and devices.

(2) - This table only applies where the above requirements are not called out in the closest military established reliability specification.

*10% lot rejection criteria for burn-in failure.
**Sealed units only.
***Parts to be manufactured on established reliability line or, burn-in of 100 hrs with max. power at 77°F (25°C).

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3.2.3 Stress analysis. The contractor shall conduct a part-level stress analysis to verify derating compliance. This analysis shall verify that all parameter deratings are compliant under the conditions of nominal unit and module power dissipation, worst case piece part stress, and worst case equipment thermal environment including worst case altitude and temperature. The contractor shall prepare a reliability stress analysis report in accordance with MIL-STD-785(see 6.2.2).

3.2.4 Parts control and standardization program. The contractor shall establish and maintain a parts selection, parts control, and parts standardization program in accordance with MIL-STD-965, Procedure I, and MIL-E-5400.The contractor shall prepare a standardization program plan(see 6.2.2). The program shall be applicable to any equipment designed or modified by the contractor or his subcontractors, but shall not apply to unmodified off-the-shelf equipment.

3.2.5 Nonstandard parts and material approval. Approval for the use of non-standard parts and materials, including electron tubes, transistors, and diodes, other than microelectronic devices, shall be obtained as specified in MIL-E-5400 and MIL-STD-965, Procedure I. The contractor shall prepare requests for approval of nonstandard parts and to proposed additions to an approved Program Parts Selection List(PPSL) (see 6.2.2). Microelectronic devices shall be approved as specified in AR-5, except that the dual source and interchangeability requirement shall not apply. Approval as a nonstandard part shall not be required for devices which meet the criteria of being listed in MIL-STD-1562, available from at least two sources which are listed in the QPL of MIL-M-38510, and procured to the requirements of MIL-M-38510 from a qualified source.

3.2.5.1 Samples required for parts and materials approval. The sample requirements for nonstandard parts and materials shall be in accordance with MIL-E-5400.

3.2.6 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of AR-5. Conformal coatings, encapsulants, embedments, or potting materials used with modular asemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.7 Modules. The electronic portions of the camera shall be modularized to the greatest extent possible in accordance with AR-10.

3.2.8 Nonrepairable subassemblies. When used, nonrepairable subassemblies shall be in accordance with MIL-E-5400 and AR-10.

3.2.9 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

3.2.10 Screw threads. Screw threads used in the construction of the camera shall be in accordance with FED-STD-H28.

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3.2.11 Locking devices and lock washers. Screw threaded assemblies subject to loosening by vibration shall be provided with locking devices or lock washers in accordance with MIL-E-5400.

3.2.12 Protective treatments. When materials are used in the construction of the camera that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, the materials shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of environmental conditions shall be prohibited.

3.3 Design. The camera shall be designed to be mounted in the Tactical Air Reconnaissance Pod System (TARPS) utilizing the KA-99( ) cradle assembly (Naval Air Development Center (NADC) Drawing No. 822-2046). The mounting locations and center of gravity of the camera shall be in accordance with Figure 1. The camera shall be designed to operate in conjunction with the AN/ASQ-172 Data Display Set, and the C-10491 Control Indicator-Power Distribution Unit. The camera design shall be in accordance with the applicable requirements of MIL-E-5400 for design and construction.

3.3.1 Weapon replaceable assemblies (WRAs). The camera shall be modular in design. The camera shall be separated into two WRAs. The camera WRAs shall be the camera body and magazine assembly (with supply and takeup cassettes installed) and the camera electronics assembly. The magazine assembly shall be capable of being removed from the camera body for the purpose of performing scheduled and unscheduled maintenance. The WRA design shall be such that any WRA, Shop Replaceable Assembly (SRA), or sub-SRA may be functionally inspected at the intermediate maintenance level independent of associated system WRAs or other weapons subsystems. All necessary adjustments, alignments, and calibration shall be accomplished at the intermediate maintenance level to eliminate the need for adjustments, alignment, and calibration at the organizational maintenance level. Suitable test points shall be incorporated into the WRA design to eliminate the need for extender devices, such as extender cards, cables, etc., at all maintenance levels. The WRA design shall include the requirements for modularization and standardization as specified in AR-10. The WRA design shall be such that fault detection and isolation shall be accomplished at the intermediate maintenance level without the need to use maintenance assist modules during repair operations. Definitions for WRA, SRA, and Sub-SRA shall be in accordance with AR-10.

3.3.2 Built-in-test (BIT). BIT provisions shall be designed to detect major voltage levels, timing, phase, input/output tolerance requirements, and film jamming. The BIT shall test the scan servo, Automatic Exposure Control (AEC), range servo, and Forward Motion Compensation (FMC). A camera failure or degraded condition detected by either automatic or manual BIT initiation shall be indicated by the use of latching indicators that shall be visible to maintenance personnel upon gaining access to the camera compartment. The design of BIT shall be such that initiation of BIT can be accomplished from either the camera or from a remote location by appropriate signal input as specified in 3.9.1.1.5.

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A failure or degraded condition in the camera shall be indicated by a 28VDC output signal for each indicator at one of the camera external connectors. The 28VDC signal shall be capable of supporting a 500 milliamperes (ma) load.

3.3.2.1 BIT features. BIT features shall eliminate the need for any auxiliary test devices or Peculiar Ground Support Equipment (PGSE) at the organizational level for the performance of WRA fault detection or fault isolation. BIT provisions shall be added to an item for the sole purpose of inspecting the item. The provisions, or circuitry, shall be simple in design and operation, accurate to within plus or minus 2 percent of the measured parameter, easily maintained, and shall not degrade the performance of the item in which the circuitry is incorporated. Failure of the BIT circuitry shall not cause a failure of the camera with which it is associated. BIT provisions shall provide optimum convenience of use and operation. The design of controls and readout devices shall be such that the controls and readout devices can be easily used and interpreted by low skill military personnel at the E-4 level. BIT provisions shall be designed to automatically provide fault information in a manner which will minimize the need for referrals to instruction and maintenance handbooks and through the self-check feature, improve the camera readiness (availability).

3.3.2.2 Self-test cycle. The BIT design shall include a self-test cycle to operate the camera for a maximum of 15 seconds to test internal functions such as film transport, scan servo operation, and AEC servo operation. The design of the self-test cycle shall be such that the self-test cycle will be activated as a part of the camera BIT sequence. Failures or degraded conditions detected during the operation of the self-test cycle shall be automatically isolated to a single WRA, and shall be indicated by activation of the appropriate latching indicator as specified herein.

3.3.3 Test points and test features. Suitable test points and test features shall be included in the design of the camera. Selection and accessibility of test points shall be compatible with the design of replaceable assemblies and the level of maintenance at which the test points are to be used.

3.3.4 Safety. The design of the camera shall incorporate the personnel safety requirements of MIL-STD-454, Requirement 1. The contractor shall prepare a safety statement, a system safety program plan, and a hazard analysis report in accordance with MIL-STD-882(see 6.2.2).

3.3.5 Human engineering. The applicable principles and criteria specified in MIL-STD-1472 shall be incorporated into the design of the camera.

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3.4 Reliability. The reliability requirements for the camera shall be 50,000 Mean Cycles Between Failures (MCBF) for the electro-mechanical portions (scanning slit mechanism and film transport) of the camera. The electronic functions and optical elements of the camera shall have a Mean Time Between Failure (MTBF) of 2,000 hours. The total camera shall have a MTBF of 117 hours. The conversion of MCBF to MTBF shall be based on a mission duration time of 2 hours and 2000 feet (609.6m) of film expended per mission.

3.4.1 Reliability program. The contractor shall establish and conduct a reliability program for the development and inspection of the camera in accordance with MIL-STD-785 and MIL-STD-781.

3.4.1.1 Reliability program plan. The contractor shall prepare a reliability program plan (production) for the camera in accordance with MIL-STD-785 (see 6.2.2).

3.4.1.2 Reliability analysis and predictions. The contractor shall prepare a reliability analysis and predictions report of the camera in accordance with MIL-STD-785 and MIL-HDBK-217(see 6.2.2).

3.4.1.3 Failure modes, effects and criticality analysis (FMECA). The contractor shall prepare a FMECA report of the camera in accordance with MIL-STD-785 and MIL-STD-2070(see 6.2.2).

3.4.1.4 Failure summary and analysis. The contractor shall prepare a failure summary and analysis report of the camera in accordance with MIL-STD-1304 (see 6.2.2).

3.4.2 Operational stability. The camera shall operate satisfactorily, continuously or intermittently, for a period of at least 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator.

3.4.3 Service life. The camera shall have a service life of at least 1,500,000 cycles when operated under any combination of environmental conditions specified herein. Allowance shall be made for corrective maintenance by replacement of worn or damaged parts.

3.5 Construction. The camera shall be constructed in such a manner that the physical and functional relationships between camera components are not affected by the strains, impacts, and vibration associated with operation, storage, and shipping. The construction of the camera shall be compatible with space limitations and shall be capable of withstanding without damage a Class 2 flight environment as specified in MIL-E-5400.

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3.5.1 Strength and rigidity of alignment. Each part of the camera shall be of sufficient strength and rigidity to acomplish its respective function and to maintain smooth operation with proper fit and precise alignment throughout the life of the camera. All parts of the drive mechanism that require timing shall be provided with timing marks.

3.5.2 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation, form, fit, function, and performance as specified in MIL-E-5400. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.5.3 Lighttightness. The camera shall be constructed to be light-tight as specified herein. The camera magazine assembly shall be lighttight to prevent the fog exposure of the film on the spool within the cassette when the magazine assembly has been removed from the camera body. The film cassette shall be lighttight to the extent that when loaded with 2000 feet (609.6m) of film having an Aerial Film Speed (AFS) of 200, no more than the first four wraps of film shall be capable of being fog exposed.

3.5.4 Fabrication. Machining, drilling, and forming shall be accomplished with the use of accurate templates, jigs and gauges. Boxes, cases, shields, and compartments shall be fabricated by casting, drawing or bending and welding, except where ease of servicing of the camera requires that a removeable panel construction be used. Where applicable stresses dictate the use of a strong aluminum or magnesium alloy that does not provide a good weld, riveting or bolting may be used.

3.6 Maintainability. The camera shall have a minimum number of parts consistent with reliability and shall be contained in a single assembly or minimum number of assemblies. The design shall utilize readily removable and replaceable units to facilitate easy assembly, disassembly, location of trouble sources and repair, with a minimum "down time" of the complete camera.

3.6.1 Mean-time-to-repair (MTTR). The MTTR at the organizational level and the intermediate level shall be as specified herein.

3.6.1.1 Organizational level. The MTTR of the camera at the organizational level shall not exceed 15 minutes. The maximum-time-to-repair shall not exceed 45 minutes for 95 percent of all corrective maintenance actions. The specified MTTR shall include the time to verify the fault/failure, the time to isolate to the faulty WRA, the time to remove and replace the faulty item, and the time required to verify the effectiveness of the repair. The time required to obtain spare parts and facilities as well as access time shall not be included. The camera shall be designed so that the alignment, adjustment, and calibration of the WRA will not be required upon replacement at this maintenance level.

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3.6.1.1.1 Detection and isolation. The camera BIT shall be capable of detecting at least 80 percent of the camera critical failures at the organizational level. A failure shall be classified critical if it results in a mission abort, no imagery or unuseable imagery. At least 90 percent of the detected critical failures shall be isolated to a single faulty WRA at the organizational level and at least 99 percent of the failure indications shall result from actual camera failures.

3.6.1.2 Intermediate level. The MTTR of the camera at the intermediate level shall not exceed 1 hour. Maximum time to repair at the intermediate level shall not exceed 2.5 hours for 95 percent of all corrective maintenance actions. The specified MTTR shall include the time to perform confirming inspections even though the removed item was identified as faulty during organizational maintenance. Alignment, adjustment, and calibration times shall be included. Time awaiting replacement parts or facilities shall not be included.

3.6.2 Test points and facilities. Test points and facilities shall be as specified herein.

3.6.2.1 Electronic test points. Electronic test points shall be in accordance with AR-10 and MIL-STD-415. When in conflict, AR-10 shall govern. All test points shall be designed to prevent overloading of the circuit under test and shall be designed so that damage to the unit shall not result from a short circuit between test points and any test point ground. Test points shall be conveniently located and shall be accessible without removal of any circuit card or component except where that test is utilized for component isolation on the circuit card. All test points shall be legibly marked.

3.6.2.2 Mechanical test points. Mechanical test points shall be provided as required and shall be in accordance with MIL-STD-415. Mechanical test points shall be simple in design and shall require minimal use of special tools.

3.6.3 Lubrication. Wherever practicable the camera shall utilize permanently lubricated sealed bearings and shall be designed to minimize requirements for routine lubrication.

3.6.4 Maintenance concept. The design of the camera shall be consistent with the maintenance concept specified herein. The camera shall be designed to be repaired at the organizational maintenance level by removal and replacement of the WRAs without the use of external test equipment. The camera shall be designed to be repaired to the bit and piece part level at the intermediate maintenance level.

3.6.4.1 Maintenance skill levels. The camera maintainability design shall permit the optical/mechanical maintenance tasks to be performed by a Navy Photographer's Mate and the electronic equipment maintenance tasks to be performed by a Navy Aviation Electronics Technician, each with the respective skill levels specified herein.

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3.6.4.1.1 Organizational maintenance skill levels. Organizational maintenance shall be performed by personnel possessing skill levels of E-4 rank and above having successfully completed the appropriate Class A and Class C Navy Technical Schooling.

3.6.4.1.2 Intermediate maintenance skill levels. Intermediate maintenance shall be performed by personnel possessing skill levels of E-5 rank and above having successfully completed the appropriate Class A and Class C Navy Technical Schooling.

3.6.4.2 Organizational level maintenance. This level of camera maintenance tasks shall be as specified herein.

3.6.4.2.1 Scheduled maintenance. Scheduled maintenance shall be limited to the removal of exposed film, the loading of unexposed film, the setting of the preflight exposure compensation (Effective Aerial Film Speed/Filter Factor Control (S/C), visual inspection of optical surfaces and film transport elements for foreign matter and cleaning when required, installation and removal of lens filters and protective cover, and performing preflight inspection by activation of BIT. No scheduled maintenance action at this level shall exceed 10 minutes.

3.6.4.2.2 Unscheduled maintenance. Unscheduled maintenance shall consist of fault detection and fault isolation to a single WRA using only BIT and removal and replacement of the faulty WRA and verification of corrective action by activation of BIT. To the maximum extent possible, the camera shall be designed so that removal and replacement of a faulty WRA can be accomplished without removing additional WRAs while the camera is installed in TARPS. No fault detection or isolation PGSE will be used at this maintenance level. Upon replacement of the WRA, no adjustment, alignment, or calibration shall be required to adapt the WRA to other camera WRAs or other camera subsystems.

3.6.4.3 Intermediate level maintenance. This level of camera maintenance tasks shall be as specified herein.

3.6.4.3.1 Scheduled maintenance. Scheduled maintenance shall include the loading of film in the magazine assembly and the periodic inspection, cleaning, and lubrication of the camera.

3.6.4.3.2 Unscheduled maintenance. Unscheduled maintenance shall consist of check out, fault detection, and fault isolation of electronic and optical/mechanical WRAs, repair of the WRAs by replacement of SRAs, and verification of Ready For Issue (RFI) condition. Repair of faulty SRAs and subassemblies will consist of fault detection, fault isolation, removal and replacement of faulty piece parts, and verification of RFI condition. Alignment and adjustment of WRAs at this maintenance level shall eliminate the need for any further alignment and adjustment when the WRA is mated with other camera components. The camera shall be Automatic Test Equipment (ATE) compatible.

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Any requirements for general purpose or common electronic test equipment or other Ground Support Equipment (GSE) shall be selected from NAVAIR 16-1-525 and NAVAIR 19-1-127. PGSE shall be limitd to adapters and interconnecting devices required to adapt the camera WRAs and SRAs to standard equipment and ATE, unless otherwise authorized by the procuring activity. All interconnecting devices shall be designed to be in compliance with MIL-STD-2077.

3.6.4.4 Depot level maintenance. Depot level maintenance shall consist of the repair of electrical, electronic, and optical/mechanical components that are beyond the capability of intermediate level maintenance.

3.6.4.5 Level of repair (LOR). The contractor shall prepare a LOR program plan, a LOR analysis report, and a LOR summary report of the camera in accordance with MIL-STD-1390 (see 6.2.2).

3.6.4.6 Extender devices. Extender devices, such as cards and adapters, shall not be used to perform normal maintenance and fault isolation at any maintenance level.

3.6.5 Maintainability program. The contractor shall establish and maintain a maintainability program for the camera in accordance with MIL-STD-470. The maintainability program shall assure that all maintenance tasks shall be capable of being perfomed by technicians within the comprehensive skill levels and maintenance times specified herein.

3.6.5.1 Maintainability program plan. The contractor shall prepare a maintainability program plan for the camera in accordance with MIL-STD-470 (see 6.2.2).

3.6.5.2 Maintainability analysis and predictions. The contractor shall prepare a maintainability analysis and predictions report of the camera in accordance with MIL-STD-470(see 6.2.2). The prediction technique shall be in accordance with MIL-HDBK-472, Procedure II, Part A, for electronic equipment and Part B, converting man-hour rate to time-to-repair, for mechanical/optical components for corrective maintenance.

3.6.5.3 Commercial equipments. The maintainability of all commercial assemblies constructed and purchased as off-the-shelf equipment shall be included in the maintainability program and shall meet the maintainability requirements specified herein.

3.7 Detail of components. The camera shall consist of the following major assemblies:

Description	Quantity
Camera Body	1 each
Camera electronics unit	1 each
Magazine assembly	1 each
Cassette	2 each

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3.7.1 Camera body. The camera body shall contain the optical barrel, camera timing mechanism, scan drive, magazine assembly, FMC mechanism, AEC mechanism, solid state recording head, and camera service life counter. The camera body shall include provisions for mounting the camera electronics unit on the camera body or in a remote location when required by the TARPS. The camera body shall provide a film format in accordance with Figure 2. The camera body shall include the wiring required to inspect the camera body and the magazine assembly as a unit.

3.7.1.1 Optical barrel. Imagery is obtained by rotation of the optical barrel which includes the scan mirror, lens, and focal plane mirror. The optical barrel shall be synchronized with the camera timing mechanism. The optical barrel shall make one revolution for each photographic cycle.

3.7.1.2 Scan and film transport drive servo. The scan and film transport drive servo shall drive the optical barrel assembly, the film transport mechanism, and shall control the rate of exposure across the scanning slit. The scan rate shall be proportional to the ground velocity divided by altitude above ground level (Vg/H) input signal during autocycle operation. The servo shall provide accurate speed control over the cycle range of 0.38 to 8 cycles per second (cps). The servo shall provide sufficient torque to accelerate the camera to 8 cps within the first 5 seconds of camera operation. Under no circumstances shall more than 20 frames of film be expended during the initial 5 second startup period.

3.7.1.3 Film transport mechanism. The film transport mechanism shall consist of suitable drive rollers, guides, sprockets, and film keeps required to properly transport a 2000 foot (609.6m) roll of 5-inch (127mm) film having a nominal base thickness of 2.5 mils and type II perforations in accordance with ANSI PH1.10. The film transport mechanism shall also be capable of properly transporting 5-inch (127mm) aerial film with a nominal base thickness of 4.0 mils and type II perforations in accordance with ANSI PH1.10. The film transport mechanism shall not generate static marks which would degrade the imagery on the processed film. The film transport mechanism shall not damage the film in any manner.

3.7.1.4 Forward motion compensation mechanism. When the camera is operating in the autocycle mode as specified in 3.9.2.1, FMC shall be accomplished by the FMC mechanism during the exposure in the appropriate direction as dictated by the design configuration of the optical path, location of the focal plane, and the direction of flight. The FMC mechanism shall be disengaged in the air-to-air mode of operation. The FMC mechanism shall automatically disengage during the pulse mode of oepration. The disengagement shall not necessarily coincide with the pulse mode switch point.

3.7.1.4.1 FMC rates. The camera shall provide FMC rates over the autocycle operating range which shall correspond to the Vg/H range of 0.08 to 1.79 radians per second (0.049 to 1.06 knots per foot).

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3.7.1.5 Automatic exposure control mechanism. The camera shall be provided with a servo system capable of accepting a Vg/H ratio signal and appropriate light sensor signal to automatically position the shutter slit width and lens diaphragm opening to provide for proper exposure of film having AFS throughout the S/C range of 3.2 to 200. The AEC mechanism shall provide automatic exposure control for all modes of camera operation throughout a scene brightness range of 12 to 8000 foot-lamberts with a Wratten 12, or equal, filter installed over the lens and when compatible with camera cycle rate and AFS rating of the film. Under conditions of decreasing luminance the AEC mechanism shall increase the aperture of the iris diaphragm to maximum before increasing the shutter slit width. Under conditions of increasing luminance, the AEC mechanism shall decrease the shutter slit width to its minimum before decreasing the aperture of the iris diaphragm. When the camera is operating in the port or starboard standoff mode, the AEC mechanism shall provide proper exposure on the film for the selected scan area.

3.7.1.5.1 Frame exposure accuracy. The exposure for any frame shall be within plus or minus 1/2 stop of the correct value except when the port or starboard standoff modes are selected. Only that portion of the exposed film corresponding to the standoff mode selected shall be within plus or minus 1/2 stop of the correct exposure value.

3.7.1.5.2 Exposure override. Provisions shall be made for exposure override. This function shall interface with the exposure override control system in the F-14 aircraft.

3.7.1.6 Camera data recording. The camera shall be capable of recording on film Binary Coded Decimal (BCD) data matrix, alpha-numeric data block, numeric frame count,center of format fiducial marks, direction of film travel mark, and flight direction mark. The BCD data matrix and the alphanumeric data block require inputs from the AN/ASQ-172 Data Display Set installed in the TARPS. The center of format fiducial marks, direction of film travel mark, and flight direction mark shall be considered fixed data. The direction of film travel mark and the flight direction mark shall be combined into the single symbol " ⌟ ", as specified in 3.7.1.6.4. Time sequencing for the numeric frame count, center of format fiducial marks, and film direction/flight direction mark shall be in accordance with Figure 2.

3.7.1.6.1 Solid state recording head. The camera shall be provided with a solid state recording head. The recording head shall be capable of being operated by the AN/ASQ-172 Data Display Set. The recording head shall be capable of being removed from and replaced into the camera. The recording head shall be capable of recording a BCD data matrix in accordance with Figure 5. The location of the data block recorded on the film shall be in accordance with Figure 2. The recording head shall be capable of recording an alphanumeric data block in accordance with Figure 6 and as specified herein. The recorded data block shall consist of 8 rows each having 8 characters. Each character shall be formed by a 5 dot by 7 dot matrix. The dimensions and location of the alphanumeric data block recorded on the film shall be in accordance with Figure 2.

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3.7.1.6.2 Numeric frame count. The camera shall be provided with a three digit resettable counter to record on the film the frame count for each frame of film as it is exposed during camera operation. The dimensions and location of the frame count shall be in accordance with Figure 2. The counter shall be reset to zero upon application of aircraft power (28VDC) to the camera. The counter shall be reset to zero upon any interruption of aircraft power to the camera that exceeds one minute.

3.7.1.6.3 Fiducial marks. The camera shall be equipped with two fiducial marks. the fiducial marks shall indicate the center of the film format along the scan direction (90 degrees). The fiducial marks shall be recorded along the top and bottom edges of the film in accordance with Figure 2.

3.7.1.6.4 Flight direction and direction of film travel marks. The camera shall record a flight direction mark and a direction of film travel mark between each frame of film. The two marks shall be combined into the single symbol " ⌟ " and shall be recorded on the film in accordance with Figure 2.

3.7.1.7 Camera cycle counter. The camera body shall contain a seven digit, additive, and nonresettable counter. The counter shall record the total number of camera cycles. The counter shall be located in a visible location on the exterior of the camera body.

3.7.1.8 Flight direction indicator. A permanently legible arrow with the words "direction of flight" shall be attached to a visible location on the exterior surface of the camera to indicate the direction of flight when the camera is mounted in the TARPS.

3.7.1.9 Lens The camera shall be provided with a 9-inch (228.6mm) ±0.136-inch (3.5mm) focal length, f/4.0 lens with sufficient field of view to prevent vignetting of the film during all modes of camera operation. The lens shall be a Type I lens in accordance with MIL-STD-150 suitable for aerial reconnaissance. The lens shall properly mount in the optical barrel.

3.7.1.9.1 Antireflection coating. All air-to-glass surfaces shall have reflection reducing coatings in accordance with MIL-C-00675.

3.7.1.9.2 Lens diaphragm. The lens shall be provided with a between the lens iris diaphragm. The diaphragm shall be capable of being automatically varied by the AEC mechanism. The diaphragm shall be calibrated from f/4.0 to f/16.0 in accordance with MIL-STD-150.

3.7.1.9.3 Lens filters. The camera shall be provided with three glass filters, Wratten 12, or equal, Wratten 25, or equal, and Clear. The filters shall be capable of being installed directly in front of the camera lens through the scan mirror viewport in the optical barrel without the use of special tools.

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3.7.1.10 Mirrors. Mirrors shall be fabricated of an ultra-low expansion material such as CER-VIT, or equal. Reflecting surfaces shall be polishd to a flatness consistent with system optical performance requirements. The mirrors shall be aluminized and overcoated with protective material in accordance with the requirements of MIL-M-13508.

3.7.1.11 Optical transmittance. The camera optical system shall have a T stop value no greater than T-5.

3.7.1.12 Optical field of view. The camera optical system shall provide a 27°20` longitudinal angle of view and shall be capable of a 180 lateral scan angle. The optical system shall provide sufficient coverage to allow for full range FMC displacement of the lens. No vignetting of the optical bundle shall occur within the specified optical field of view during any mode of camera operation.

3.7.1.13 Electronics and electromechanical devices. The camera body shall be provided with a power supply, electronics circuitry, and electromechanical devices to perform Vg/H signal conditioning, timing and sequencing, scan and film drive servo, AEC compensation including aperture and slit width positioning, range focus positioning, data recording, and fiducial mark imaging.

3.7.2 Camera electronics unit. The camera electronics unit shall interface the camera with the TARPS power and control signals. The camera electronics unit shall provide the signals required for the numeric frame count, center of format fiducial marks, direction of film travel mark, and flight direction mark. The camera electronics unit shall contain the electronics circuitry and controls for the camera. The camera electronics unit shall contain a time totalizing meter in accordance with MIL-M-7793. The S/C switch, BIT initiate switch, BIT indicators, and test points (via test connector) shall be located in the camera electronics unit. All of the control devices shall be modularized to the greatest extent possible. Removable circuit boards shall be utilized wherever possible. Semiconductor devices shall be utilized to the greatest extent possible. The camera electronics unit shall be capable of being mounted on the camera body or remotely as required by the TARPS.

3.7.3 Magazine assembly. The camera shall be provided with a film magazine as a part of the magazine asembly. The film magazine shall contain the film transport mechanism. The magazine shall provide the necessary seating surfaces and fasteners for mounting the two cassettes to the magazine and for mounting the magazine to the camera body. The magazine shall be of sufficient

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strength and rigidity to maintain proper alignment throughout the life of the camera. Mechanical and electrical connections required between the magazine and the camera body shall be automatically engaged or disengaged when the components are mated or separated, respectively. Mechanical connections shall include all mechanical linkages such as gears, shafts, interlocks, and other mechanical drive and control elements.

3.7.3.1 Film loading. The design of the magazine assembly shall facilitate the threading of the film through the magazine when the magazine is detached from the camera body. The design of the magazine shall permit the attachment of new film in the supply cassette to the film remaining in the magazine. Suitable means shall be provided in the magazine assembly to aid in the alignment of the two films and expedite the splicing operation.

3.7.3.2 Magazine counter. The magazine shall contain a seven digit, additive, and nonresettable counter. The counter shall be located in a visible location on the exterior of the magazine. The counter shall record the total number of camera operation cycles when the magazine is installed on the camera body.

3.7.4 Film cassettes. The camera shall be provided with one supply cassette and one takeup cassette as part of the magazine assembly. The cassettes shall be identical and completely interchangeable with each other. All mechanical and elecrical connections shall automatically engage when the cassettes are mated to the camera magazine. The camera magazine shall be provided with permanently legible indicators that indicate the correct position for the supply cassette and the takeup cassette. These indicators shall be visible when the camera is installed in the TARPS. The cassettes shall be lighttight to the extent specified in 3.5.3. A data card, 3 inches (76.2mm) wide by 1.5 inches (38.1mm) high, shall be provided on each cassette for making pencil notations. The data card shall be constructed from material that allows the pencil notations to be easily erased.

3.7.4.1 Film capacity. Each cassette shall have a capacity of 2000 feet (609.6m) of 5-inch (127mm) film having a nominal base thickness of 2.5 mils and wound on a MS-26565-12 spool.

3.7.4.2 Film supply indicator. Each cassette shall be equipped with a film supply indicator. The indicator shall show the amount of film remaining in the cassette. The indicator shall be in a convenient visible location on the exterior of the cassette.

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3.7.4.3 End-of-film sensor. The magazine shall be provided with an end-of-film sensor. The sensor shall be capable of stopping camera operation at the maximum camera cycle rate when there is not more than 30 feet (9.1m) of film remaining in the supply cassette. The sensor shall provide an end-of-film signal as specified in 3.9.1.2.2.

3.7.5 Dust covers. The camera body and magazine assembly shall be provided with easily removable and replaceable dust covers which shall protect mechanisms and optics.

3.8 Environmental conditions. The camera shall operate satisfactorily under the conditions specified herein.

3.8.1 Sand and dust. The contractor shall prepare an environmental report (see 6.2.2), to verify that the camera will withstand sand and dust particles encountered under service conditions in operational areas of the world when installed in the TARPS.

3.8.2 Fungus. The contractor shall prepare an environmental report(see 6.2.2), to verify that approved non-nutrient materials are used throughout the camera to the greatest extent possible and that all other materials used shall be suitably treated to resist fungus growth as encountered in tropical climates.

3.8.3 Attitude. The camera shall operate in any attitude or position.

3.8.4 Temperature-attitude. The camera shall meet the temperature-altitude requirements specified herein.

3.8.4.1 Equipment operating. The camera shall operate properly under all combinations of temperature and altitude in the ranges of 0^{0}F to 120^{0}F (-17.8^{0}C to 48.9^{0}C) and sea level to 50,000 feet (15,240m). When operated at 0^{0}F(-17.8^{0}C) the camera shall be permitted to use internal heaters for a period of ten minutes prior to operation.

3.8.4.2 Equipment nonoperating. The camera shall withstand all combinations of temperature and altitude in the ranges of -65^{0}F to 160^{0}F (-53.9^{0}C to 71.1^{0}C) and sea level to 50,000 feet (15,240m).

3.8.5 Humidity. The camera shall withstand the effects of humidities up to 100 percent, including conditions where condensation takes place in and on the camera except as specified herein.

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3.8.6 Salt atmosphere. The camera shall withstand exposure to salt sea atmosphere under service conditions encountered in naval operations.

3.8.7 Explosive atmosphere. The camera shall be capable of operating in an explosive atmosphere. The camera shall not cause the ignition of an ambient explosive gaseous mixture with air when operated in such an atmosphere.

3.8.8 Acceleration. The camera shall meet the acceleration requirements specified herein.

3.8.8.1 Equipment operating. The camera shall withstand the steady state stresses due to acceleration while operating in the TARPS on the F-14 aircraft. The camera performance during operation shall not be degraded and malfunctions shall not occur while flying the TARPS designated mission profiles for the F-14 aircraft.

3.8.8.2 Equipment nonoperating. The camera shall withstand the acceleration loads as calculated for the flight envelope of the F-14 aircraft.

3.8.9 Vibration. The camera shall not be damaged and shall operate properly when mounted with vibration isolators in place and subjected to vibrations in the range of 20 to 2000 Hertz (HZ) with the power spectral density as specified in Figure 3.

3.8.10 Shock. The camera with vibration isolators in place shall not be damaged and shall meet the performance requirements of Section 3 when subjected to impact shock of 15g's. The camera mounting base with excursion stops or bumpers in place, individual isolators, other attaching devices, and maximum rated load applied in a normal manner shall withstand impact shock of 30g's. The camera shall be capable of withstanding the shock environment encountered during servicing operatings.

3.9 Performance characteristics. The camera shall perform as specified herein.

3.9.1 Operation. The warmup time required for the camera prior to operation under service conditions shall not exceed 10 minutes. The camera shall require no warmup time prior to operation under room ambient conditions. The camera shall require no special cooling. The camera shall not be required to operate when the temperature of the camera exceed 150°F (65.6°C) while installed in the TARPS. No special preflight thermal sabilization shall be required for proper operation of the camera. Built-in heaters may be used to provide heating during aircraft engine startup and during the mission as required. No more than 10 minutes of preheating time shall be required prior to camera operation. The camera, when

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supplied with electrical power and command signals, shall be capable of operating in an autocycle mode at 0.38 to 8 cps with a 55 percent overlap and in a pulse mode at a fixed scanning rate of not less than 0.50 cps, at a cycling rate determined by an externally provided intervalometer signal that is less than, or equal to, 0.40 pulse per second.

3.9.1.1 Input signals. Input signals shall be provided to the camera as specified herein.

3.9.1.1.1 Vg/H signal. A positive DC voltage proportional to Vg/H and appropriately scaled to cover all modes of camera operation shall be provided. The scale factor in the normal mode shall be 50.3(volts per knot per foot). When operating in either standoff mode, the scaled voltage shall be sin 30^0 times 50.3(volts per knot per foot) or 25.15(volts per knot per foot) where 30^0 represents the angle from the horizon where the overlap from frame to frame shall be 55 percent. The scale change to standoff shall be made external to the camera. For 8 cps, the maximum operational voltage shall be 53.7 volts. A return line shall be provided for the Vg/H voltage. Both the signal line and the return line shall have resistive loads greater than 100,000 ohms.

3.9.1.1.2 Cloud cover exposure signals. Two 28VDC cloud cover exposure signals which provide a +1 and a +2 stop exposure increase override shall be provided.

3.9.1.1.3 Vehicle altitude signal. A positive voltage proportional to the vehicle altitude (H) shall be provided. The scale factor shall be 0.5 volts per 1000 feet(304.8m) Above Ground Level (AGL). The maximum H shall be 70,000 feet(21,336m) (35 volts). The load resistance shall be greater than 100,000 ohms. A return line shall be provided for the H voltage. When the loss of altitude signal voltage occurs, the camera shall automatically self-focus to an altitude of 6,000 feet (1,829m) AGL.

3.9.1.1.4 Intervalometer pulse train signal. An intervalometer pulse train signal shall be provided. The pulse rate shall be appropriately scaled to cover all modes of camera operation. The pulse source shall provide a pulse rate of 7.5 Vg/H pulses per second where Vg/H is scaled as a function of the selected mode as specified in 3.9.1.1.1. The pulse amplitude shall be +28 volts and the pulse width shall be 75 plus or minus 25 milliseconds.

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3.9.1.1.5 BIT initiate signal. A BIT initiate signal shall be provided. The signal shall be a momentary switch closure to 28VDC with a minumum pulse width of 10 milliseconds.

3.9.1.1.6 Interface with LED RHA. The Light Emitting Diode Recording Head Assembly (LED RHA) at the camera shall be comprised of a linear, 1x60 LED array positioned so that sucessive firings of the LEDs shall produce a BCD data matrix or an alphanumeric data matrix in accordance with Figure 5 and Figure 6, respectively. The placement and orientation of data blocks in the camera film format between frames of photographic images shall be in accordance with Figure 2. The input signals to the LED RHA are in accordance with Figure 7 as the LED drive portion of AN/ASQ-172 Data Display Set - camera interconnection block diagram. One set of output signals from a single electrical connector, 1J4 on the AN/ASQ-172 Data Display Set, is routed directly to the camera LED RHA connector J3, through assigned wires in the interconnecting cable in accordance with Figure 8. Connector pins and associated input functions for the J3 connector shall be as specified in 3.9.1.3.

3.9.1.1.6.1. LED RHA drive signals. The AN/ASQ-172 Data Display Set shall provide the required pulses to drive the linear, 1x60 LED array. The AN/ASQ-172 Data Display Set shall be capable of firing all 60 LEDs simultaneously. Each data laydown shall consist of a maximum of 60 firings repeated a maximum of 40 times. The data laydown shall be all BCD, all alphanumeric, or alternate as determined by the setting of the appropriate controls of the AN/ASQ-172 Data Display Set. A recommended LED firing current pulse is shown in Figure 9. Any shape pulse which supplies an amount of energy equivalent to the recommended pulse within the same adjustable time period shall be acceptable, provided that the maximum current does not exceed 825ma and provided that the annotation is successfully accomplished. The maximum allowable diode voltage drop shall be 4.0 volts. The LED drive circuits shall be designed to be fail safe. Regardless of the type of failure, current to any LED shall not remain on for more than 50 μsec. Interface circuitry for LED firing shall be in accordance with Figure 10.

3.9.1.1.6.2 BCD form. The BCD form shall be accomplished by firing LED rows 5 to 36. A total of 18 firings shall be employed to produce a data matrix.

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3.9.1.1.6.3 Alphanumeric form. The alphanumeric form shall be accomplished by firing all 60 LED rows. A total of 40 firings shall be employed to produce a data matrix. Each alphanumeric character shall be formed by firing 7 contiguous LEDs a maximum of five times. Definition of the character line is shown below. LED rows 40, 48, and 56 shall not be utilized. The AN/ASQ-172 Data Display Set is connected to a total of 60 rows.

Line	LED Rows
8	1 through 7
7	9 through 15
6	17 through 23
5	25 through 31
4	33 through 39
3	41 through 47
2	49 through 55
1	57 through 63

3.9.1.1.7 Camera command signals. The signals specified herein shall consist of 28VDC closures.

3.9.1.1.7.1 Ready command signals. When the signal is de-energized (OFF), the heaters, BIT, and interlock open functions shall be operative. When the signal is energized (ON), all camera circuitry shall become operative except the scan and film transport system.

3.9.1.1.7.2. Operate command signal. When the signal is energized (ON), the scan and film transport functions shall operate providing the photographic performance of the camera system. The heaters and blowers shall be de-energized (OFF) during camera operation.

3.9.1.1.7.3 Air-to-air command signal. When the signal is energized (ON), the camera shall operate at a constant 1 cps with the range focus set at infinity.

3.9.1.1.7.4 Port standoff command signal. When the signal is energized (ON), the AEC shall respond to the port photosensor only. The Vg/H signal voltage shall be scaled as specified in 3.9.1.1.1. The camera shall operate at one-half the normal auto-cycle rate.

3.9.1.1.7.5 Starboard standoff command signal. When the signal is energized (ON), the AEC shall respond to the starboard photosensor only. The Vg/H signal voltage shall be scaled as specified in 3.9.1.1.1. The camera cycle rate shall be as specified in 3.9.1.1.7.4.

3.9.1.2 Camera output signals. The camera shall provide 28VDC electrical output signals as specified herein.

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3.9.1.2.1 Film remaining signal. The camera shall provide a 28VDC pulse for each camera cycle. The pulse shall drive a remote frames remaining counter and shall energize a remote 28VDC 500ma lamp load. The 28VDC pulse duration shall be equal to one-half of the camera operate cycle duration. The minimum pulse duration shall be 62 milliseconds at 8 cps.

3.9.1.2.2 Film failure signal. The camera shall contain a film failure switch with circuitry which shall transmit a steady state signal to a remote fail indicator when a film jam, film breakage, or expenditure of film occurs. The signal shall stop camera operation. The signal shall be removed by placing the ready command signal (3.9.1.1.7.1), in the "OFF" position.

3.9.1.2.3 Interlock open signal. The signal shall be a 28VDC switch closure capable of supporting a 500ma lamp load. The signal shall indicate end of film supply, lack of film supply, takeup cassette or supply cassette not properly latched to the magazine, magazine connectors or electronics unit connectors not mated to the camera body, or the remote cable is not properly connected when the electronics unit is remotely located.

3.9.1.2.4 WRA failure indicator signals. The camera shall provide a 28VDC pulse of 75 milliseconds minimum duration. The pulse shall be capable of supporting a 500ma lamp load for each WRA failure. When failures have occurred, the WRA failure indicator signal output(s) shall occur following the initiation of BIT (3.9.1.1.5).

3.9.1.2.5 BIT failure signal. The camera shall provide a remote 28VDC pulse of 75 milliseconds minimum duration capable of supporting a 500ma lamp load when a camera failure occurs following the initiation of BIT (3.9.1.1.5). The signal shall activate a magnetic latching "NO-GO" indicator located on the electronics unit of the camera.

3.9.1.2.6 Data signals. The data signals shall be as specified herein.

3.9.1.2.6.1 Data freeze signal. The data freeze signal shall command the AN/ASQ-172 Data Display Set to stop updating information and begin formatting the required BCD or alphanumeric outputs. The data freeze signal and its relationship to the other timing signals shall be in accordance with Figure 11. Interface circuitry shall be in accordance with Figure 12.

3.9.1.2.6.2 Data print signal. The data print signal shall command the AN/ASQ-172 Data Display Set to begin data output to the camera. The data print signal and its relationship to the other timing signals shall be in accordance with Figure 11. Interface circuitry shall be in accordace with Figure 12.

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3.9.1.2.6.3 File spacing signal. The file spacing signal shall define 0.018-inch (0.46mm) of film motion to permit proper spacing of the data block columns. The file spacing signal and its relationship to other timing signals shall be in accordance with Figure 11. Interface circuitry shall be in accordance with Figure 12.

3.9.1.2.6.4 Film sensitivity control signal. The film sensitivity control signal shall be a mission constant DC voltage determined by the camera S/C switch setting. The DC voltage amplitude shall determine the required current from the AN/ASQ-172 Data Display Set to the LED RHA installed in the camera. The required LED current, as a function of film sensitivity control voltage, shall be in accordance with Figure 13. Interface circuitry shall be in accordace with Figure 14.

3.9.1.3 Interface connectors. Signal and power interface connector J1 Pin designation(s) and function(s) shall be as specified in Table II. The camera interface connector J3 Pin designation(s) and function(s) shall be as specified in Table III.

TABLE II. J1 connector pin/function designations.

Connector Pin	Function
1,2, and 3	A/C 115VØA
4,5, and 6	A/C 115VØB
7,8, and 9	A/C 115VØC
10, and 11	A/C 28VDC
12, and 13	Power Ground
17	Ready Command
18	Operate Command
19	Port Standoff Mode
20	Starboard Standoff Mode
21	Air-To-Air Mode
22	Intervalometer
23	BIT Initiate
24	AEC + 1 Stop
25	AEC + 2 Stops
32	BIT Fail Indicator Flag
33	Frames Remaining
34	Film/Fail Indicator
35	Interlock Open Indicator
36	Body Fail Indicator Flag
37	Magazine Fail Indicator Flag
38	Electronics Unit Fail Indicator Flag
47	Vg/H Signal Hi
49	Altitude Signal Hi
50	Altitude Signal Return

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TABLE III J3 connector pin/function designations.

Connector Pin	Function	Connector Pin	Function
1	Data Freeze +	43	Rows 25 - 30 Cathode
2	Data Freeze -	44	LED Row 31 Anode
3	Data Print +	45	LED Row 32 Anode
4	Data Print -	46	LED Row 33 Anode
5	File Spacing +	47	LED Row 34 Anode
6	File Spacing -	48	LED Row 35 Anode
7	Film Sensitivity +	49	LED Row 36 Anode
8	Chassis (Gnd)	50	Rows 31 - 36 Cathode
9	LED Row 1 Anode	51	LED Row 37 Anode
10	LED Row 2 Anode	52	LED Row 38 Anode
11	LED Row 3 Anode	53	LED Row 39 Anode
12	LED Row 4 Anode	54	LED Row 41 Anode
13	LED Row 5 Anode	55	LED Row 42 Anode
14	LED Row 6 Anode	56	LED Row 43 Anode
15	Rows 1 - 6 Cathode	57	Rows 37 -43 Cathode
16	LED Row 7 Anode	58	LED Row 44 Anode
17	LED Row 8 Anode	59	LED Row 45 Anode
18	LED Row 9 Anode	60	LED Row 46 Anode
19	LED Row 10 Anode	61	LED Row 47 Anode
20	LED Row 11 Anode	62	LED Row 49 Anode
21	LED Row 12 Anode	63	LED Row 50 Anode
22	Rows 7 - 12 Cathode	64	Rows 44 - 50 Cathode
23	LED Row 13 Anode	65	LED Row 51 Anode
24	LED Row 14 Anode	66	LED Row 52 Anode
25	LED Row 15 Anode	67	LED Row 53 Anode
26	LED Row 16 Anode	68	LED Row 54 Anode
27	LED Row 17 Anode	69	LED Row 55 Anode
28	LED Row 18 Anode	70	LED Row 57 Anode
29	Rows 13 - 18 Cathode	71	Rows 51 - 57 Cathode
30	LED Row 19 Anode	72	LED Row 58 Anode
31	LED Row 20 Anode	73	LED Row 59 Anode
32	LED Row 21 Anode	74	LED Row 60 Anode
33	LED Row 22 Anode	75	LED Row 61 Anode
34	LED Row 23 Anode	76	LED Row 62 Anode
35	LED Row 24 Anode	77	LED Row 63 Anode
36	Rows 19 - 24 Cathode	78	Rows 58 - 63 Cathode
37	LED Row 25 Anode	79	Chassis (Gnd)
38	LED Row 26 Anode		
39	LED Row 27 Anode		
40	LED Row 28 Anode		
41	LED Row 29 Anode		
42	LED Row 30 Anode		

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3.9.2 Modes of operation. The camera shall be capable of operating in the normal mode, the port standoff mode, the starboard standoff mode, and the air-to-air mode in response to remote commands.

3.9.2.1 Normal mode. When in the normal mode, the camera shall be capable of operating in autocycle or pulse. Autocycle operation shall provide a continuous and uniform cycling rate that is proportional to the vehicle supplied Vg/H signal specified in 3.9.1.1.1. FMC shall be provided. Pulse operation shall provide a single, fixed speed camera cycle for each vehicle supplied intervalometer pulse specified in 3.9.1.1.4. The transition between autocycle and pulse operation, as well as FMC engaged/disengaged, shall be internally controlled. The camera shall respond to the proper H signal while in the normal mode of operation.

3.9.2.2 Standoff mode. When in the port standoff mode and the starboard standoff mode, the camera shall operate as specified in the normal mode, except that the driving signals shall be externally rescaled by a constant factor so as to provide stereo overlap of 55 percent at an oblique depression angle of 30 degrees.

3.9.2.3 Air-to-air mode. When in the air-to-air mode, the camera shall operate in autocycle at a fixed rate of 1 cps. FMC shall be disengaged and the slant range servo shall be driven to the infinity focus position.

3.9.3 Continuous range focus correction. The camera shall be capable of continuously correcting for range focus error over an altitude range of 500 feet (152.4m) to infinity. An externally supplied altitude signal shall be provided to the range servo so that range focus correction may be accomplished.

3.9.4 Relative shutter speeds. The camera relative shutter speed ranges shall be 1/43 to 1/1070 second at 0.38 cps and 1/900 to 1/22,600 second at 8 cps.

3.9.5 Camera system resolution. The camera positioned at infinity focus, loaded with EK3412 (PAN-X) film, a Wratten No. 12, or equal, filter over the lens, and utilizing high contrast (1000:1) resolution targets in accordance with MIL-STD-150, shall provide the minimum average dynamic bench resolution in the image area as specified herein.

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3.9.5.1 FMC disengaged. With FMC disengaged, the camera shall provide the following minimum average dynamic bench resolution in the image area:

Mode and cycle rate (cps)	Shutter speed (sec)	Resolution (lp/mm)
Autocycle		
0.50	1/1000	80
3	1/2000	75
8	1/2000	60
Pulse		
0.50	1/1000	75

3.9.5.2 FMC engaged. With FMC engaged, the camera shall provide the following minimum average dynamic bench resolution in the image area:

Mode and cycle rate (cps)	Shutter speed (sec)	Resolution (lp/mm)
Autocycle		
0.50	1/1000	71
3	1/2000	67
8	1/2000	53

3.10 Electrical requirements. The camera shall operate from AC and DC power as specified herein.

3.10.1 Input power. The camera shall meet all applicable requirements of MIL-STD-704 and shall be capable of providing the specified performance from power sources with characteristics as specified in MIL-STD-704. The contractor shall specify the effects of voltage transients on camera performance in accordance with MIL-STD-704. The input power limitations are as follows:

(a) AC power (three phase) 115/200 volt 400 Hz
Camera system steady state - 1450 watts maximum
Heaters - 400 watts maximum
Unbalance limits for 3 Ø power shall be as specified in MIL-STD-704.

(b) DC power
Operating voltage limits - 24 to 28.5 volts,
category B steady state - 50 watts maximum

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3.10.1.1 Undervoltage protection. The camera shall not be damaged when subjected to the minimum voltages specified in Tables I and II of MIL-STD-704. The camera shall automatically resume normal operation when the voltage returns to within normal limits.

3.10.1.2 Electrical overload protection. Each internal regulated power supply in the camera shall be of the current-limiting type, and shall be capable of continuous operation at any load condition between no load and short circuit conditions. The voltage of a regulated supply shall not decrease for currents up to 130 percent of the maximum current used from the supply. When an overload condition ceases to exist for a regulated power supply, the supply shall automatically resume normal regulation within 0.5 second. Each internal unregulated power supply shall be protected against damage from current overload conditions.

3.10.2 Electrical interface connectors. Electrical interface connections between the camera body and magazine assembly, and the camera body and electronics unit shall be made through connectors in the mating surfaces that shall make proper electrical contact when the mating surfaces are engaged. The electrical interface connectors shall meet the requirements of MIL-C-24308 and shall be in accordance with Figure 15. The design of the connectors shall be such that reverse and improper assembly, mating, and installation is not possible.

3.10.3 Cabling and connections. Cabling and connections shall be as specified herein.

3.10.3.1 Interconnecting cabling. The camera shall meet the requirements of this specification when operated with external wiring in accordance with the applicable requirements of MIL-W-5088. Airplane cabling between the camera and other airplane systems shall be as provided by the procuring activity. The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when required to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. Interconnecting signals shall be differential and shall not depend on mounting structure for signal transmission.

3.10.3.2 Wire connectors. Wire connectors shall be utilized to the greatest extent possible to preclude the cutting or unsoldering of wires when dismantling the camera, replacing major subassemblies, and reassembling the camera. Connectors shall be keyed to prevent the improper insertion of a connector or subassembly into the wrong location within or on the camera.

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3.10.3.3 External electrical connectors. External electrical connectors shall be selected in accordance with MIL-STD-1353 and shall be in accordance with Figure 15.

3.10.4 Circuit identification. Circuits shall be identified in accordance with MIL-STD-681. Circuit tracing throughout the camera shall be accomplished without the use of auxiliary means.

3.11 Electromagnetic interference. The camera shall meet the electromagnetic interference requirements of MIL-STD-461 for Class IC equipment. The camera shall meet the narrow band requirements of MIL-STD-461 and shall demonstrate that the narrow band levels are less than the levels which would occur if the rated current load of the camera were being drawn by a resistive load, whichever is higher.

3.12 Dimensions. The maximum overall dimensions of the camera, excluding the cradle assembly specified in 3.3, shall not exceed 36 inches (914.4mm) in length, 20.25 inches (514.4mm) in width and 18.38 inches (466.9mm) in height.

3.13 Weight. The weight of the camera loaded with 2000 feet (609.6m) of 5-inch (127mm) film, excluding cables and the cradle assembly specified in 3.3, shall not exceed 240 pounds (108.9kg).

3.14 Finish. The exterior and interior finishes of the camera shall be as specified herein.

3.14.1 Exterior. All exterior surfaces of the camera, except as specified herein, shall have a corrosion and abrasion resistant finish in accordance with MIL-E-5558, type I wrinkle-tinted. The color shall be in accordance with FED-STD-595, Number 16473. Machine surfaces used for the purpose of accurately mounting parts shall be left in their smooth state. Machine surfaces shall be protected by an electrolytic or chemical finish when feasible. Except when prevention of radio interference is required, the machine surfaces shall be protected by a light application of high grade, hard industrial wax.

3.14.2 Interior. All interior surfaces of the camera shall be given a nonreflecting, dull, and durable black finish wherever practicable. The provisions of 3.14.1 pertaining to machine surfaces shall apply.

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3.15 Identification of product. The camera shall be marked for identification in accordance with MIL-STD-130.

3.15.1 Nomenclature and nameplates. Nomenclature assignments and nameplate approval for camera identification shall be in accordance with MIL-N-18307.

3.15.2 Serial number. A serial number shall be assigned to each major component. The serial number shall not be duplicated for any similar piece of equipment.

3.15.3 Stock number. Upon request of the contractor, the stock number shall be supplied by the contracting officer.

3.16 Type designator. The parenthesis in the type designator will be replaced by a suffix letter and a production issue number, in accordance with the provisions of MIL-STD-155. This complete type designator shall then be used on nameplates, shipping records, in instruction books, and wherever reference is made to the specific equipment defined by the designator. As soon as possible after award of the contract, the contractor should apply to the contracting officer for such information.

3.17 Drawings. When specified by the procuring activity (see 6.2), a complete set of level 3 drawings and all associated lists as specified in DOD-D-1000 shall be prepared (see 6.2.2).

3.17.1 Microfilm. When specified by the procuring activity (see 6.2), microfilm records of the drawings and associated data lists shall be prepared as specified in MIL-M-9868(see 6.2.2).

3.18 Container. A durable container shall be provided to protect the camera and accessories during shipment and storage.

3.19 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.5).
2. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the inspection conditions specified herein.

4.3.1 Temperature and humidity. Unless otherwise specified, inspections shall be conducted in an ambient temperature of 77°F (25°C), plus or minus 5 degrees, with any relative humidity up to 90 percent.

4.3.2 Input power. The inspections shall be conducted with nominal 28VDC and nominal 115VAC, 400Hz, single phase power sources.

4.4 Failure during inspection. Any characteristics exhibited by the camera such as deterioration, power loss, sticking, excessive power drain, corrosion, etc., during any of the inspections specified herein which would cause disability of the camera to meet any of the requirements of Section 3, while in service, shall constitute failure of the camera to pass the inspection and may be the basis for rejection of the camera.

4.4.1 Corrective validation. When a failure occurs, the individual inspection requirements schedule shall include an inspection to check all cameras for the noted nonconformance until determination has been made that the defect has been satisfactorily corrected.

4.5 First article inspection. The first article inspection shall be as specified herein.

4.5.1 First article samples. The first article samples shall consist of models representative of the production camera. Unless otherwise specified by the procuring activity, the schedule for the first article inspection shall be in accordance with Table IV.

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TABLE IV. First article inspection samples.

Inspection	Camera No. 1	Camera No. 2	Camera No. 3
Sand and dust		X	
Fungus		X	
Attitude			X
Temperature-altitude		X	
Humidity		X	
Explosive atmosphere	X		
Acceleration (equipment operating)	X		
Acceleration (equipment nonoperating)	X		
Vibration	X		
Shock	X		
Electromagnetic interference		X	
Salt fog		X	
Reliability demonstration	X		X
Maintainability demonstration			X

4.5.2 Environmental inspections.

4.5.2.1 Sand and dust. When specified by the procuring activity, (see 6.2), the camera shall be inspected in accordance with Method 510.1, Procedure I of MIL-STD-810.

4.5.2.2 Fungus. When specified by the procuring activity (see 6.2), the camera shall be inspected in accordance with Method 508.1, Procedure I of MIL-STD-810.

4.5.2.3 Attitude. The camera, loaded with 2000 feet (609.6m) of 5-inch (127mm) film, shall be operated at a cycling rate of 1 cps for a period of 2 minutes in each of six positions. After operation, the camera and film shall be inspected to determine that the camera and film have sustained no damage as a result of operating in each of the six positions.

4.5.2.4 Temperature-altitude. The camera shall be inspected in accordance with Method 504.1 of MIL-STD-810 for equipment category 5, except as specified herein. The steps of Table 504.1-II shall be amended as follows:

Step 1b - The chamber temperature shall be adjusted to -65°F (-53.9°C).
Step 2 - The chamber temperature shall be adjusted to 0°F (-17.8°C).
Step 3 - The chamber temperature shall be adjusted to 0°F (-17.8°C).
Step 6 - The chamber temperature shall be adjusted to 160°F (71.1°C).
Step 8 - The chamber temperature shall be adjusted to 150°F (65.6°C).

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4.5.2.5 Humidity. The camera shall be inspected in accordance with Method 507.1, Procedure I of MIL-STD-810. At the conclusion of the 10th cycle of step 6 and within 1 hour after removal of the camera from the chamber, the camera shall be operated to determine that the camera performs as specified in Section 3.

4.5.2.6 Salt fog. The camera shall be inspected in accordance with Method 509.1, Procedure I of MIL-STD-810 except that the chamber temperature shall be maintained at 75°F(23.9°C) and the inspection duration shall be 24 hours.

4.5.2.7 Explosive atmosphere. The camera shall be inspected in accordance with Method 511.1, Procedure I of MIL-STD-810 using 100/130 octane gasoline. Step 1 shall be performed at 120° F (48.9°C).

4.5.2.8 Acceleration (equipment operating). The camera shall be inspected in accordance with Method 513.2, Procedure II of MIL-STD-810. The vehicle category for Table 513.2-II shall be aircraft carried stores.

4.5.2.9 Acceleration (equipment nonoperating). The camera shall be inspected in accordance with Method 513.2, Procedure I of MIL-STD-810. The vehicle category for Table 513.2-I shall be aircraft carried stores.

4.5.2.10 Vibration. The camera shall be inspected in accordance with Method 514.2, Procedure IIA of MIL-STD-810 except that the endurance level for the camera shall be $16.53g_{rms}$ and the function level shall be $9.65g_{rms}$. The inspection duration shall be 1 hour functional and 1 hour endurance. The inspection sequence shall be the functional inspection performed for 0.5 hour, followed by 1 hour of the endurance inspection, followed by the final 0.5 hour of the functional inspection. The vibration inspection curves, which describe the test levels for three orthogonal axes of the camera, are in accordance with Figure 3.

4.5.2.11 Shock. The camera shall be inspected in accordance with Method 516.2, procedures I, III, and V of MIL-STD-810. Procedure V shall apply only to the camera electronics unit. The shock pulse shape for Procedures I and III shall be in accordance with Figure 516.2-2 of MIL-STD-810.

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4.5.2.12 Electromagnetic interference. The camera shall be inspected in accordance with the test methods of MIL-STD-462 specified for Class IC equipment of MIL-STD-461. The test method RS03 frequency range shall be 14 kilohertz (KHz) to 20 gigahertz (GHz) at 200 volts/meter.

4.5.3 Reliability demonstration. The camera shall be inspected in accordance with Test Plan XIIIC of MIL-STD-781 except as specified herein. The environmental inspection level shall be as specified in MIL-STD-781 for air lauched weapons and external stores except that the temperature range shall be 0^{0}F to 120^{0}F (-17.8^{0}C to 48.9^{0}C). Test Plan XIIIC of MIL-STD-781 shall be modified to convert hours to cycles in accordance with 3.4.

4.5.4 Maintainability demonstration. The camera shall be inspected for scheduled maintenance and unscheduled maintenance at the organizational level and unscheduled maintenance at the intermediate level. The inspection shall be performed in accordance with Method 9 of MIL-STD-471 at the 95th percentile level for scheduled maintenance at the organizational level and Method 8 of MIL-STD-471 for unscheduled maintenance at both the organizational and intermediate levels.

4.6 Quality conformance inspection. Each camera submitted for acceptance on the contract or order shall have passed the quality conformance inspections specified herein.

4.6.1 Physical examination. Examine each camera visually and with gauges to determine conformance to specified requirements for materials, construction, fit and interchangeability of components, dimensions, weight, finish, markings, wiring, color coding and other visible defects. Operate all controls, releases, locks, and moving parts to determine suitability for the intended purpose. Examine each camera for ease of attachment and detachment of unit parts.

4.6.2 Operational. Each camera shall be inspected in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2), to determine that the camera meets the specified requirements of 3.3, 3.7, 3.9, and 3.10 for BIT, film format, optical barrel, scan and film transport drive servo, film transport mechanism, camera cycle counter, electronics and electromechanical devices, camera electronics unit, magazine counter, film footage indicator, end-of-film sensor, warmup time, cycling rates, input signals, camera output signals, modes of operation, and input power during live film operation at various camera cycle rates.

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4.6.3 Camera lighttightness. Each camera, assembled as a complete unit and loaded with unexposed film having and AFS of 200, shall be exposed to the illumination of a 17,800 lumen No. 2 photoflood lamp on each of the six sides of the camera. The photoflood lamp shall be placed at a distance of five feet from the camera. The total time of exposure to the illumination of the photoflood lamp shall be thirty minutes divided equally between the six sides of the camera. Upon completion of the procedure, the film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected for evidence of a light leak in the camera. Evidence of a light leak shall be cause for rejection.

4.6.4 Cassette lighttightness. Each cassette, loaded with unexposed film having an AFS of 200, shall be exposed to the illumination of a 17,800 lumen No. 2 photoflood lamp on each of the cassettes four sides. The four sides of the cassettes shall be the two ends, and two nonoverlapping 180 degree sections of the circumference of the cassette. The photoflood lamp shall be placed at a distance of five feet from the cassette. The total time of exposure to the illumination of the photoflood lamp shall be twenty minutes divided equally between the four sides of the cassette. During exposure to one section of the circumference of the cassette, the illumination of the photoflood lamp shall be directed at the cassette light trap. Upon completion of the procedure, the film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected for evidence of fog exposure. Evidence of fog exposure beyond the first four wraps of film shall be cause for rejection.

4.6.5 Forward motion compensation. Each camera shall be inspected to determine conformance to the FMC requirements of 3.7.1.4, and 3.9.2.1. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2), except as specified herein. A resolution inspection with FMC engaged shall be performed. The camera shall be loaded with unexposed film having an AFS of 40. The FMC mechanism shall be engaged and the camera shall be operated at cycling rates of 0.50, 3, and 8 cps. The radial and tangential resolution values shall be combined in accordance with the expression $\sqrt{RT}$, where R=line pairs/ mm for the radial portion of the resolution target and T=line pairs/mm for the tangential portion of the resolution target. The average degradation in resolution shall not exceed one pattern over the non-FMC resolution inspection result.

4.6.6 Automatic exposure control. Each camera shall be inspected to determine conformance to the requirements of 3.7.1.5 for the shutter and lens diaphragm to provide the proper exposure by responding to AEC functions for controlled inputs of brightness, cycling rate, and S/C settings. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2), except as specified herein. Suitable calibrated light sources shall be utilized in the inspection procedure. The AEC shall be inspected in three

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modes of operation; normal, starboard stand-off and port standoff. When in the standoff mode of operation, only the photocell viewing the standoff position selected shall be energized. When in the normal mode of operation, both the port and starboard photocells shall be energized to average scene brightness. When measuring shutter slit response, each of four brightness levels (50, 500, 2500, and 5000 foot-lamberts) shall be applied to the photocell(s) and the response of the shutter slit measured and recorded. The camera shall be operated at each of four cycle rates (0.38, 1, 4, and 8 cps) and the response of the shutter slit shall be measured and recorded. When measuring the response of the lens diaphragm for a given scene brightness level, the S/C switch shall be set a each of four positions (4, 20, 100, and 200) and the response of the diaphragm measured and recorded.

4.6.7 Camera data recording. Each camera shall be inspected to determine conformance to the requirements of 3.7.1.6. The camera shall be loaded with unexposed EK3414 film. The camera shall be operated in conjunction with the AN/ASQ-172 Data Display Set or other equipment that is capable of simulating the specified data inputs to the camera. The exposed film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected to determine that the format, dimensions, and locations of the alphanumeric data block, BCD data block, numeric frame count, center of format fiducial marks, and direction of flight/direction of film travel marks are in accordance with Figure 2.

4.6.8 Lens. Each lens shall be inspected to determine conformance to the requirements of 3.7.1.9 and 3.7.1.9.2. The inspection shall be performed in accordance with the applicable methods of MIL-STD-150 and procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

4.6.9 Antireflection coating. Each lens shall be inspected in accordance with the procedures of MIL-C-00675.

4.6.10 Filters. Each filter shall be inspected in accordance with the procedures of MIL-STD-1278.

4.6.11 Mirrors. Each mirror shall be inspected for spectral reflection characteristics and figures in accordance with the procedures of MIL-M-13508.

4.6.12 Optical system. Each complete optical system shall be inspected to determine conformance to the requirements of 3.7.1.11 and 3.7.1.12. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

4.6.13 Continuous range focus correction. Each camera shall be inspected to determine conformance to the requirements of 3.9.3. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

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4.6.14 Relative shutter speeds. Each camera shall be inspected to determine conformance to the requirements of 3.9.4. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

4.6.15 System resolution. Each camera shall be inspected to determine conformance to the requirements of 3.9.5. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

4.6.16 Reliability assurance. Each camera shall be inspected in accordance with Test Plan XVIIIC of MIL-STD-781 for 100 hours minimum, with θ_1 equal to 58.5 hours. The environmental inspection level shall be as specified in MIL-STD-781 for air launched weapons and external stores except that the temperature range shall be $0^{o}F$ to $120^{o}F$ ($-17.8^{o}C$ to $48.9^{o}C$). The inspection shall be truncated at 200 hours operating time for each camera. This inspection applies to the electronic and optical portions of the camera. The electromechanical portion of the camera may be exercised with dummy film. Temperature cycling inspections shall be in accordance with MIL-STD-781 except that the temperature range shall be $0^{o}F$ to $120^{o}F$ ($-17.8^{o}C$ to $48.9^{o}C$). During this inspection the camera shall be subjected to a functional inspection at both the beginning and prior to the end of each operating portion of the temperature cycle. Prior to the 100-hour inspection of each camera, a burn-in period may be used at the option of the contractor. When used, the burn-in period shall be the same for each camera in time and severity in accordance with MIL-STD-781. Failure criteria and definitions shall be as specified in MIL-STD-2074. The contractor shall prepare reliability test and demonstration reports in accordance with MIL-STD-781 (see 6.2.2).

4.6.17 Undervoltage protection. Each camera shall be inspected to determine conformance to the requirements of 3.10.1.1. The inspection shall be performed in accordance with procedures prepared by the contractor and approved by the procuring activity (see 6.2.2).

4.6.18 Electrical overload protection. Each camera shall be inspected to determine that the internal regulated power supply(s) will handle all load conditions between no load and a short circuit by imposing a short circuit on the power supply(s). Measurements shall be taken and recorded to determine that the regulated power supply(s) is capable of resuming normal regulated operation within 0.5 second after removal of the short circuit.

Failure of the regulated power supply to do so shall be cause for rejection. Additional voltage measurements shall be taken to assure that no decrease in the regulated power supply(s) voltage occurs for current values fo 130 percent of the required maximum operating current.

4.7 Rejection and retest. Rejected cameras shall be replaced or reworked to correct the defects, after which all necessary inspections shall be repeated. Rejected cameras shall not be resubmitted for inspection without preparing failure reports (see 6.2.2). When investigation indicates that the defects may exist in cameras, previously accepted, failure reports shall be prepared (see 6.2.2).

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4.8 Destroyed specimen. All parts, specimens, or assemblies destroyed in making inspections required to determine compliance with this specification, shall be in addition to the quantity specified in the contract and shall be furnished without increasing the cost of the contract or order.

4.9 Packaging inspection. Examine the packaging and packing of the camera in accordance with the Quality Assurance Provisions of the specifications referenced in Section 5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or Commercial as specified by the procuring activity (see 6.2).

5.1.1 Level A. The camera shall be preserved and packaged in accordance with Method IIB of MIL-P-116. Selection and application of preservative shall be at the option of the contractor in accordance with the applicable provisions of MIL-P-116.

5.1.2 Commercial. The camera shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A or Commercial as specified by the procuring activity (see 6.2).

5.2.1 Level A. The camera, packaged as specified in 5.1.1, shall be packed in a class 2, style 2 wooden box in accordance with PPP-B-621.

5.2.2 Commercial. The camera shall be packed in a shipping container in a manner that will afford adequate protection against

damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract, each unit package, and intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The camera covered by this specification is intended for use in low and medium altitude reconnaissance photography.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Number of cameras required.
(c) First article samples required (see 3.1).
(d) Drawings required (see 3.17).
(e) Microfilm required (see 3.17.1).
(f) Sand and dust inspection required (see 4.5.2.1).
(g) Fungus inspection required (see 4.5.2.2).
(h) Level of packaging and packing required (see Section 5).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664)and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.3	Report, Reliability Stress Analysis	UDI-R-21423
3.2.4	Contractor Standardization Program Plan	UDI-P-21282A
3.2.5	Nonstandard & Parts Approval Request/Proposed Additions to An Approved PPSL	DI-E-7028
3.3.4	Safety Statement System Safety Program Plan Hazard Analysis Report	DI-H-1322A DI-H-1320A DI-H-3278
3.4.1.1	Reliability Program Plan (Production)	DI-R-5129A
3.4.1.2/ 3.6.5.2	Report, Reliability and Maintainability Prediction	UDI-R-21134
3.4.1.3	Report, Analysis, Failure Modes and Effects	UDI-R-21140
3.4.1.4	Report, Failure Summary and Analysis	DI-R-7041

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3.6.4.5	Report, Level of Repair Summary Plan	DI-L-2082A
	Level of Repair Program Report	DI-R-2084
	Level of Repair Analysis	DI-L-2085A
3.6.5.1	Plan, Maintainability Program	DI-R-2127
3.8.1/3.8.2	Report, Environmental	UDI-R-21138
3.17/3.17.1	Drawings, Engineering and Associated Lists	DI-E-7031
4.6.2/4.6.5/ 4.6.6/4.6.8/ 4.6.12/4.6.13/ 4.6.14/4.6.15/ 4.6.17	Test Procedures	UDI-T-21347
4.6.16	Reports, Reliability Test and Demonstration	DI-R-7034
4.7	Reports, Failure	UDI-R-21141A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directe by the contracting officer).

6.3 First article. When a first article is required for inspection and approval (see 3.1), the first article sample shall consist of three cameras (see 4.5.1).

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract. The contract shall have precedence over any specification.
(b) This specification. This specification shall have precedence over all applicable subsidiary specifications.
(c) Referenced specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Preparing activity:
Navy-AS

(Project No. 6720-N304)

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NOTES:

1. DIMENSIONS ARE IN INCHES WITH MILLIMETER EQUIVALENTS IN PARENTHESES.
2. TOLERANCES ARE ±0.02 FOR 2 PLACE DECIMALS AND ±0.005 FOR 3 PLACE DECIMALS UNLESS OTHERWISE NOTED.

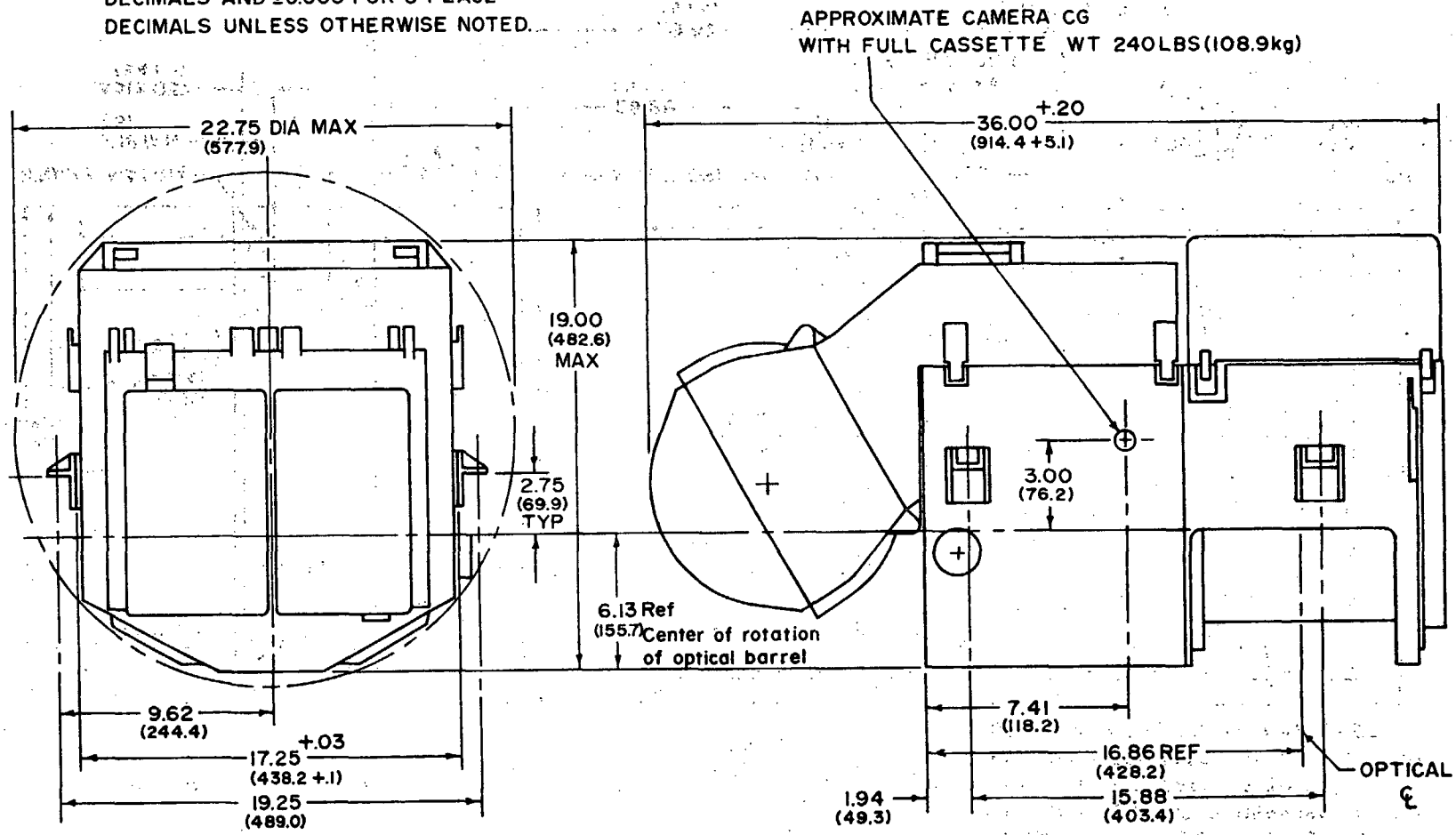


FIGURE 1. KA-99 Panoramic Camera Outline.

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NOTES:

1. DIMENSIONS ARE IN INCHES WITH MILLIMETER EQUIVALENTS IN PARENTHESES.
2. TOLERANCES ARE ±0.02 FOR 2 PLACE DECIMALS AND ±0.005 FOR 3 PLACE DECIMALS UNLESS OTHERWISE NOTED.
3. DIMENSIONS REFERRING TO DATA BETWEEN FORMATS ARE BASED UPON DOT CENTERS AND/OR ROW LINES.

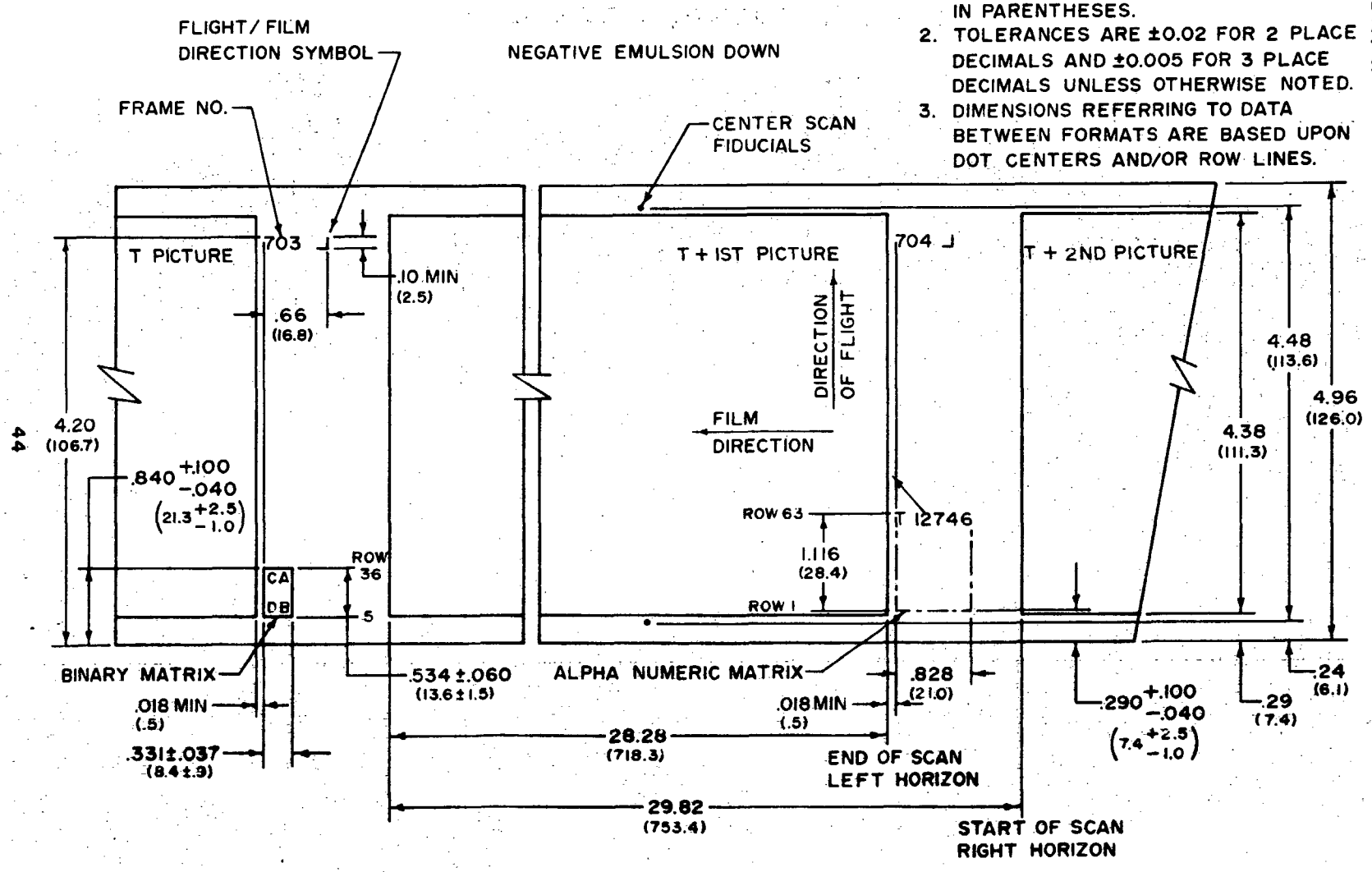
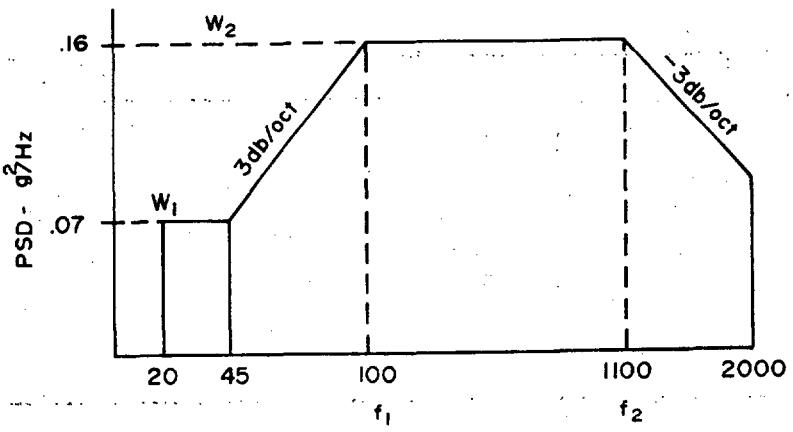


FIGURE 2. Film Format.

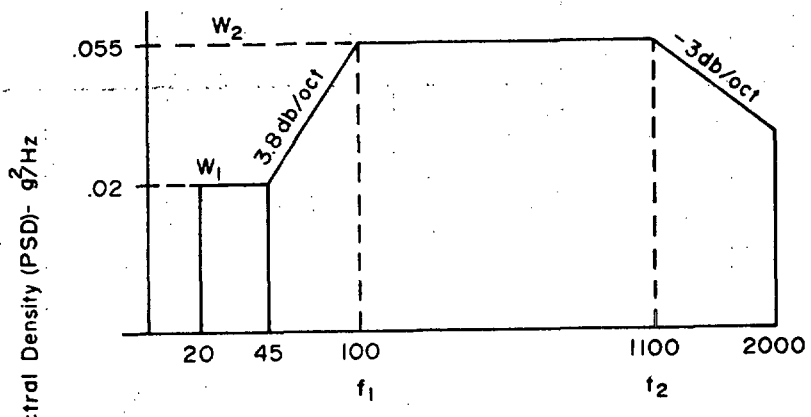
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ENDURANCE LEVEL



grms = 16.53

FUNCTIONAL LEVEL



grms = 9.65

FIGURE 3. Vibration Spectrum.

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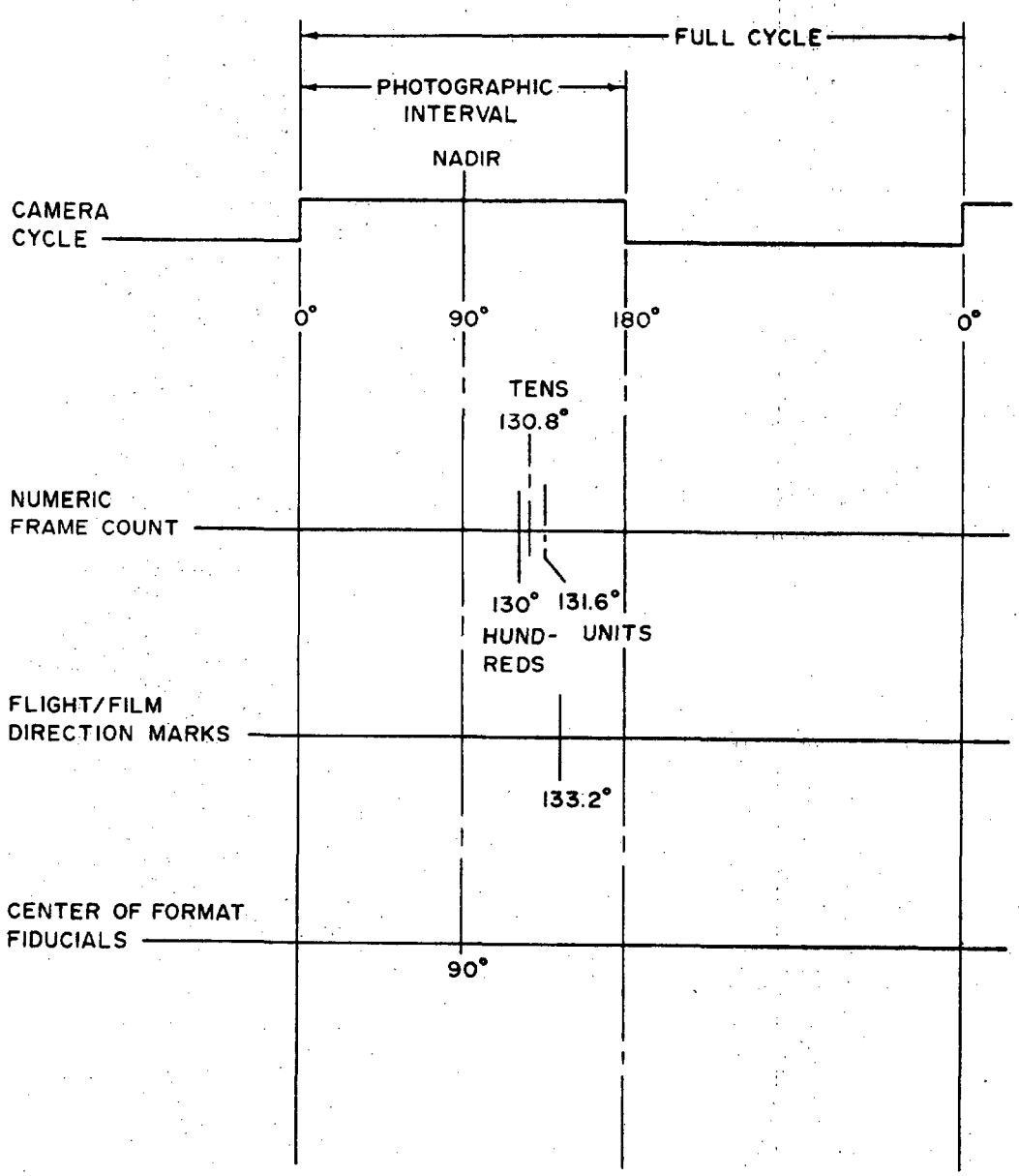
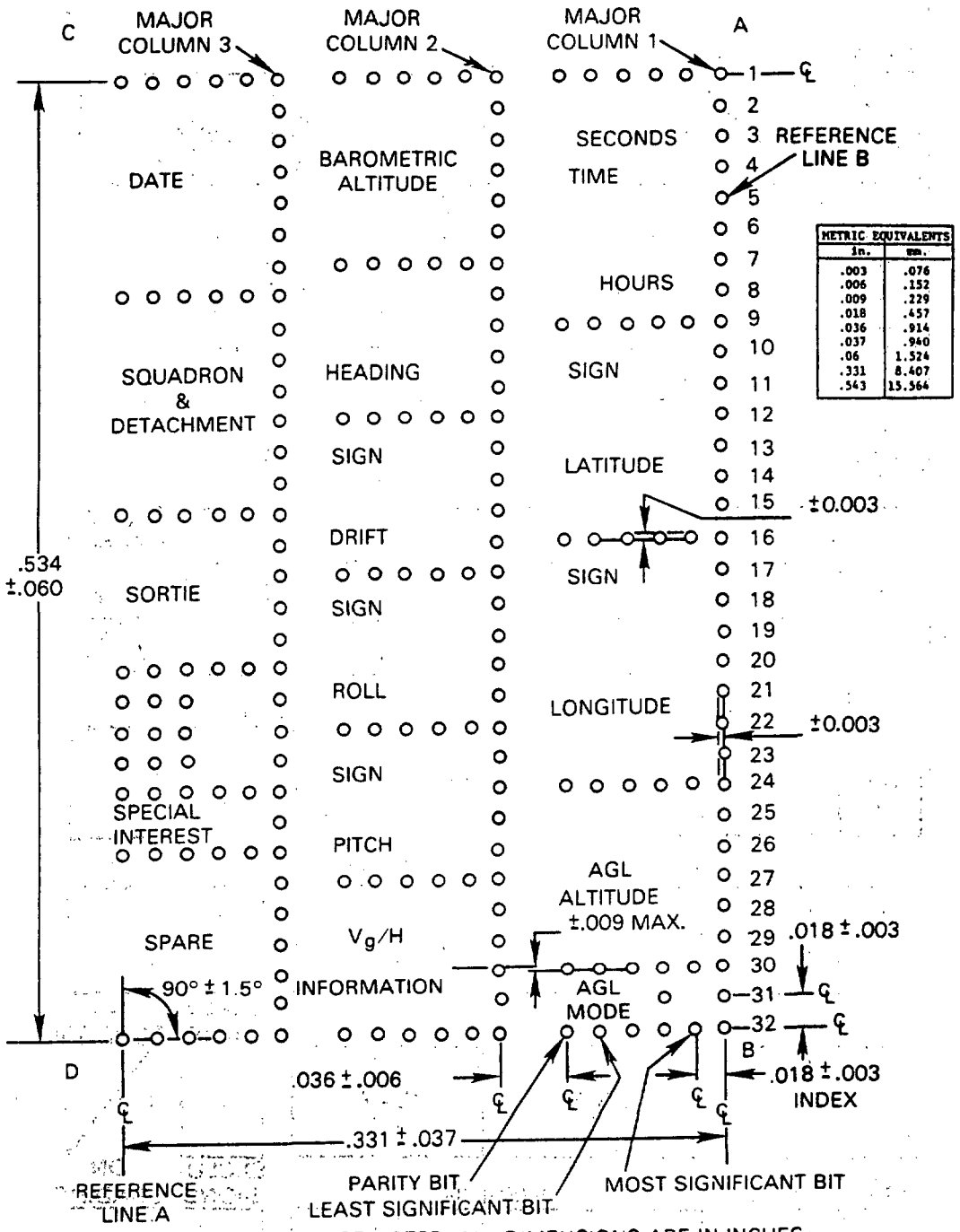


FIGURE 4. Timing Sequence For Numeric Frame Count And Fixed Data.

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NOTE: UNLESS OTHERWISE NOTED, ALL DIMENSIONS ARE IN INCHES.

FIGURE 5. BCD Data Matrix Format.

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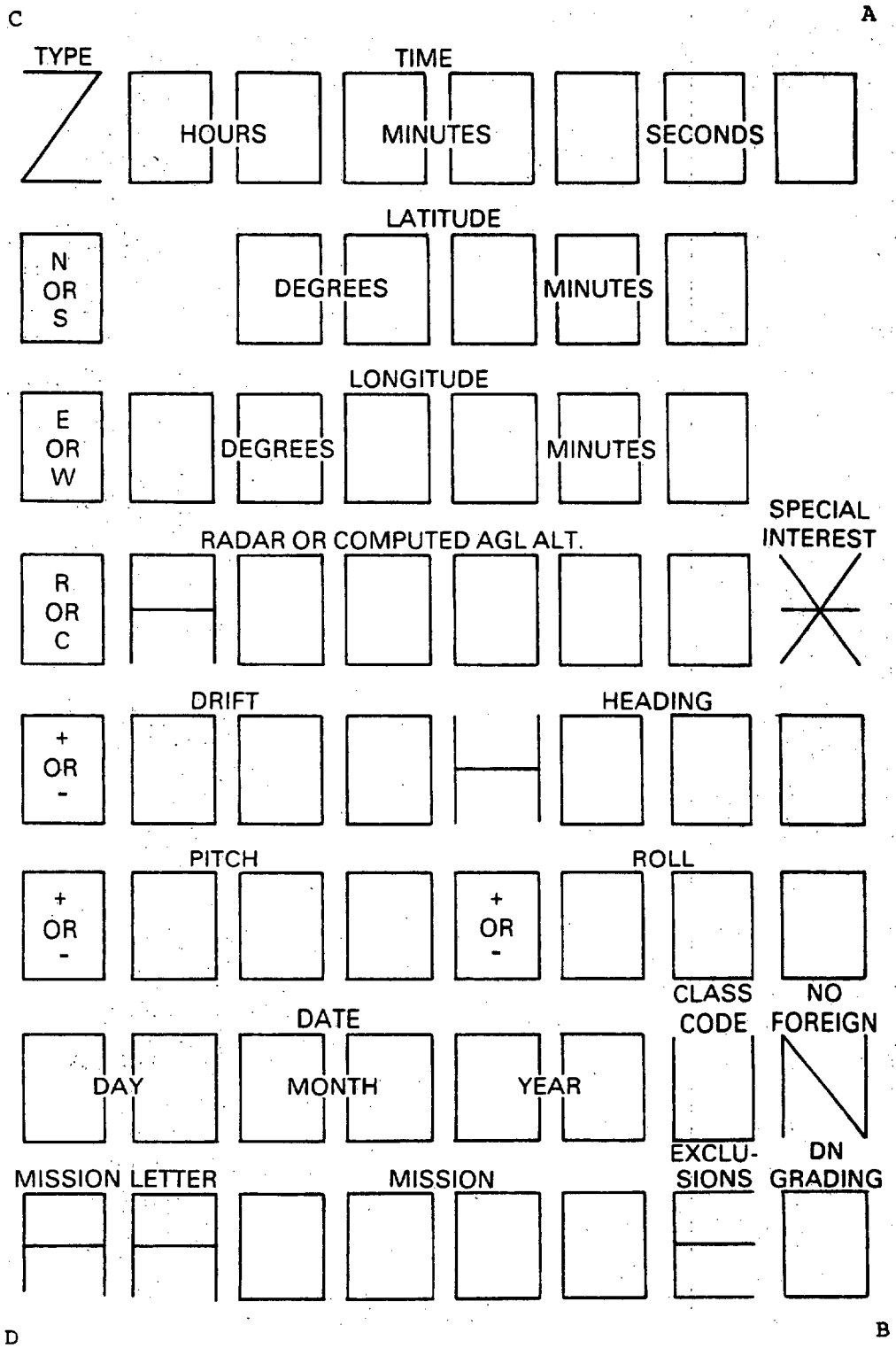


FIGURE 6: A-N Data Matrix Format.

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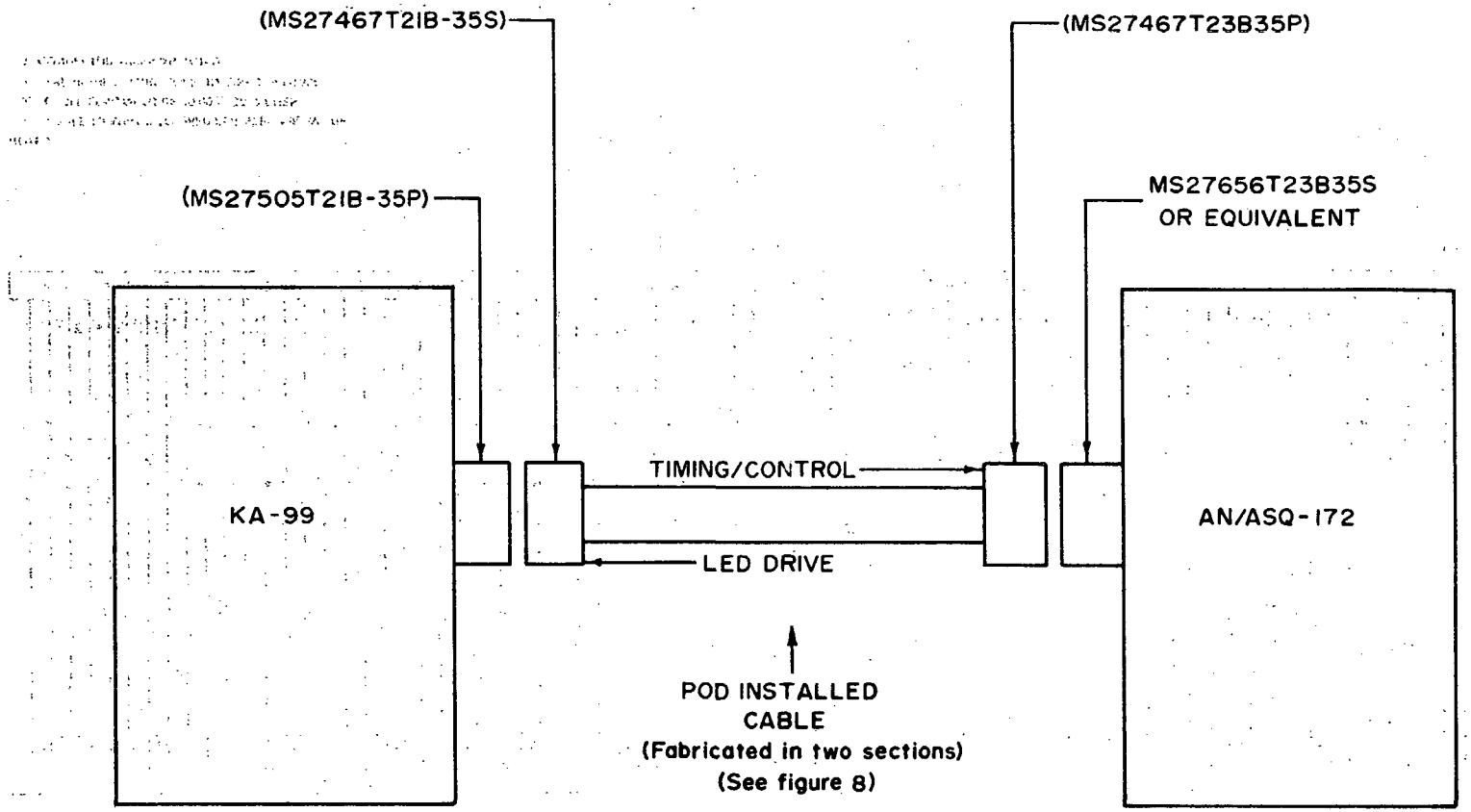
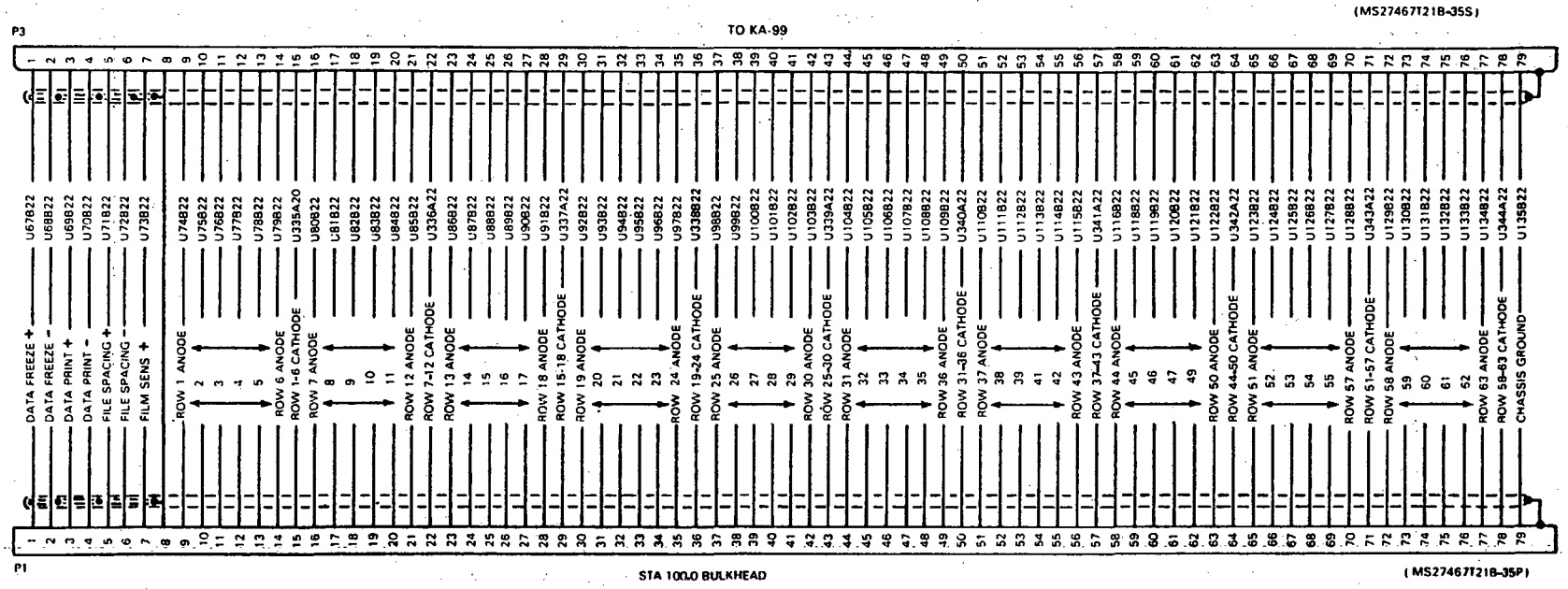


FIGURE 7. KA-99 - AN/ASQ - 172 Interconnection Block Diagram.

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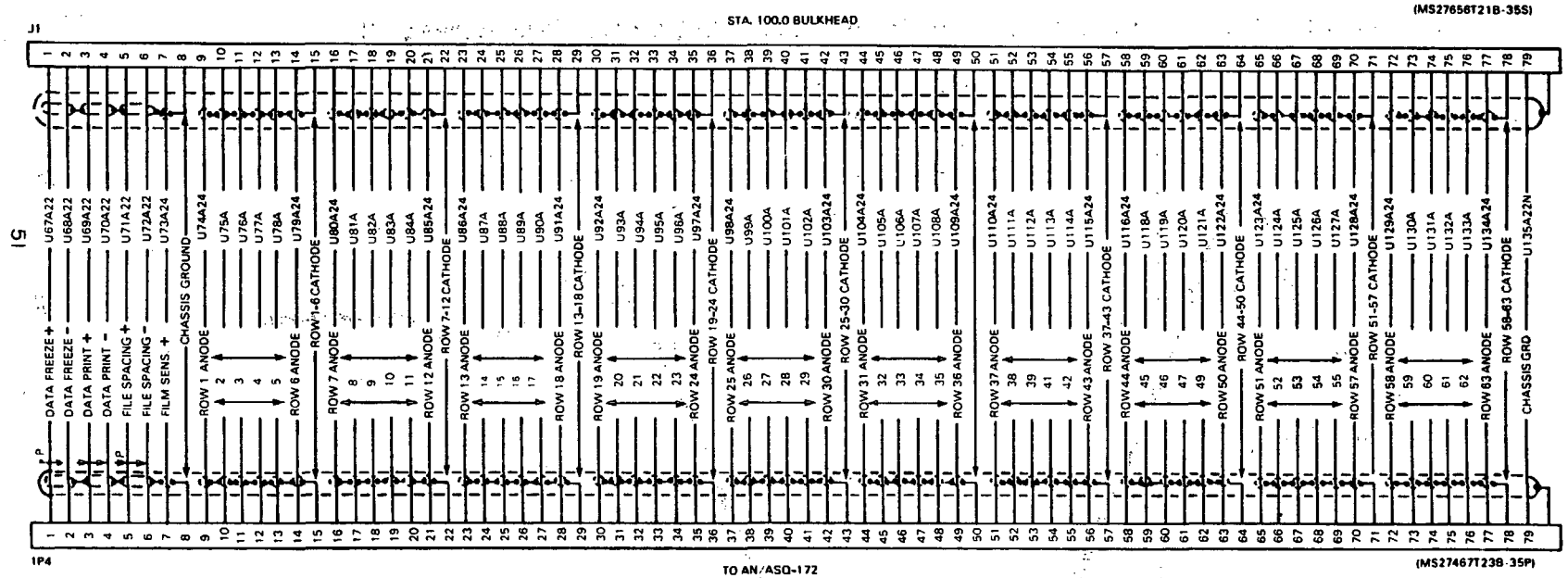


NOTES:

1. CABLE TO MANUFACTURED PER SPEC MIL-W-5088
2. CABLE LENGTH TO BE 30.00 ± .50 INCHES
3. THE HIGHEST WIRE CODE NO. USED IS U135B
4. CONNECTOR TYPES AS NOTED

FIGURE 8. KA-99 - AN/ASQ-172 Interconnecting Cable Definition.

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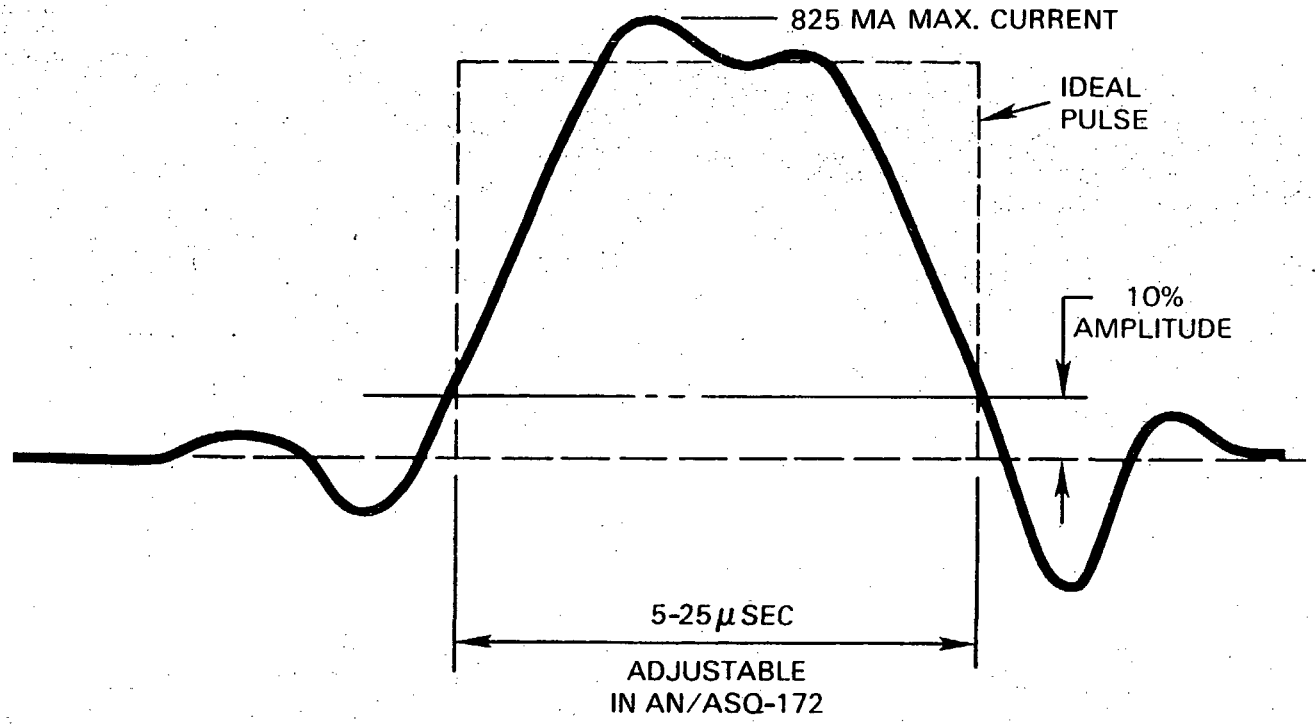


NOTES:

1. CONNECTOR TYPES: AS NOTED
2. CABLE TO BE MANUFACTURED PER SPEC MIL-W-5088
3. CABLE LENGTH TO BE 132.00 ± .50 INCHES
4. THE HIGHEST WIRE CODE NO. USED IS U135A

FIGURE 8. KA-99-AN/ASQ-172 Interconnecting Cable Definition. - Continued.

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1. TYPICAL PULSE FIRING INTO LED THROUGH NINE FT. OF CABLE DEFINED IN FIGURE 6.
2. LED FORWARD DROP 4.0 V.MAX @750 MA
3. PULSE SHAPE MAY DEVIATE FROM ABOVE SO LONG AS EQUIVALENT ENERGY IS CONTAINED WITHIN THE REAL ENVELOPE.

FIGURE 9. Recommended LED Firing Pulse.

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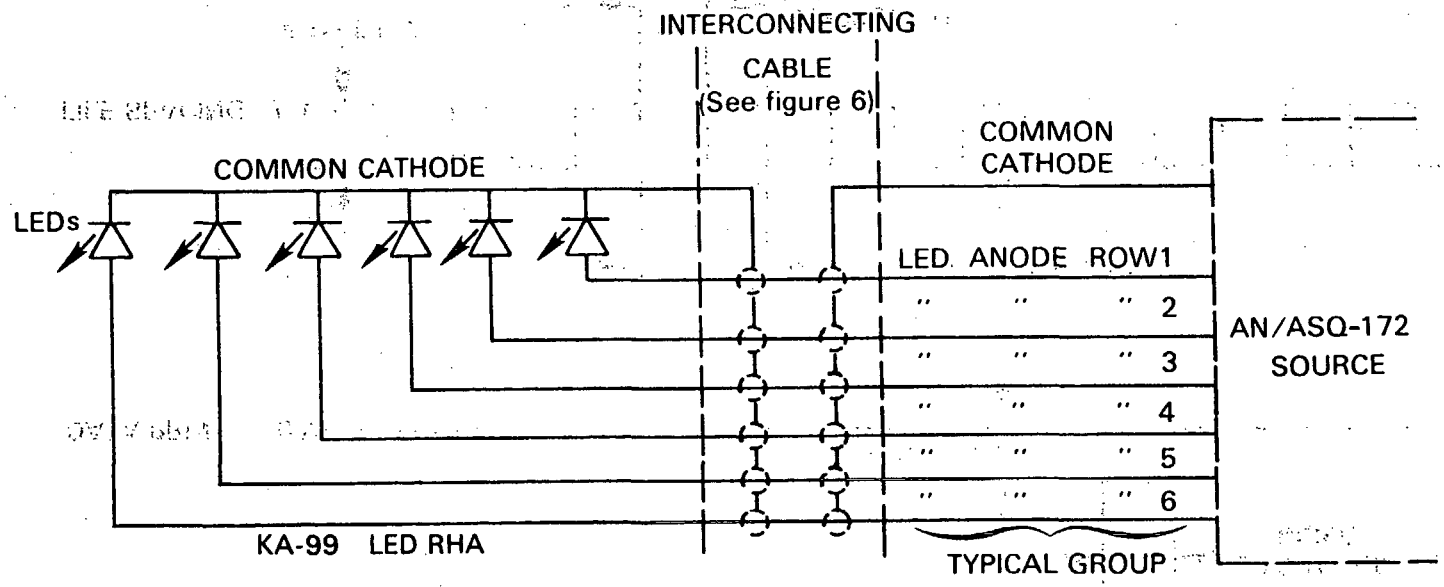
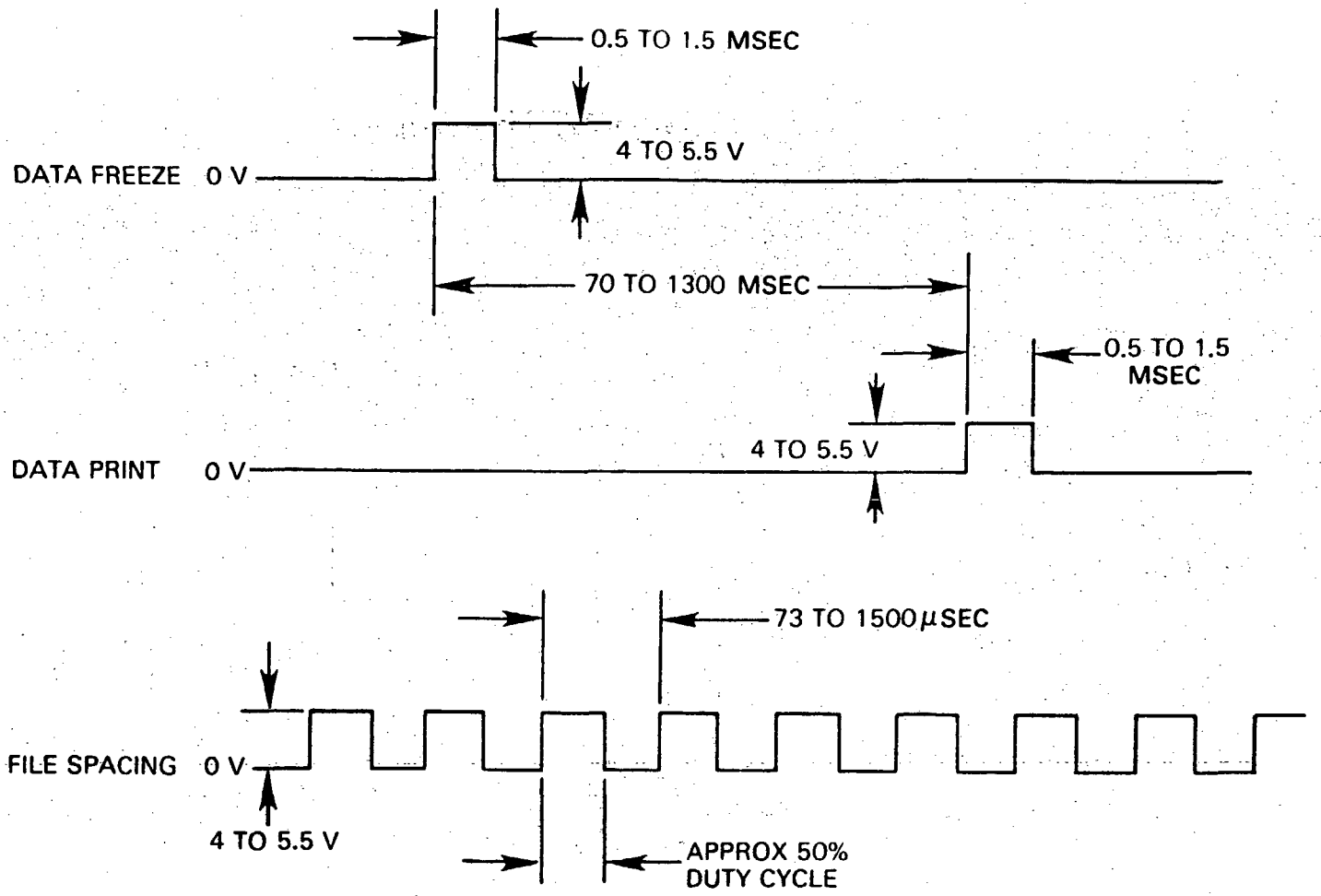


FIGURE 10. LED Firing Circuit.

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NOTE: When the KA-99 is operating, a minimum of ten (10) cycles of the file spacing signal will occur prior to the first occurrence of data freeze.

FIGURE 11. KA-99 Output Timing Signals.

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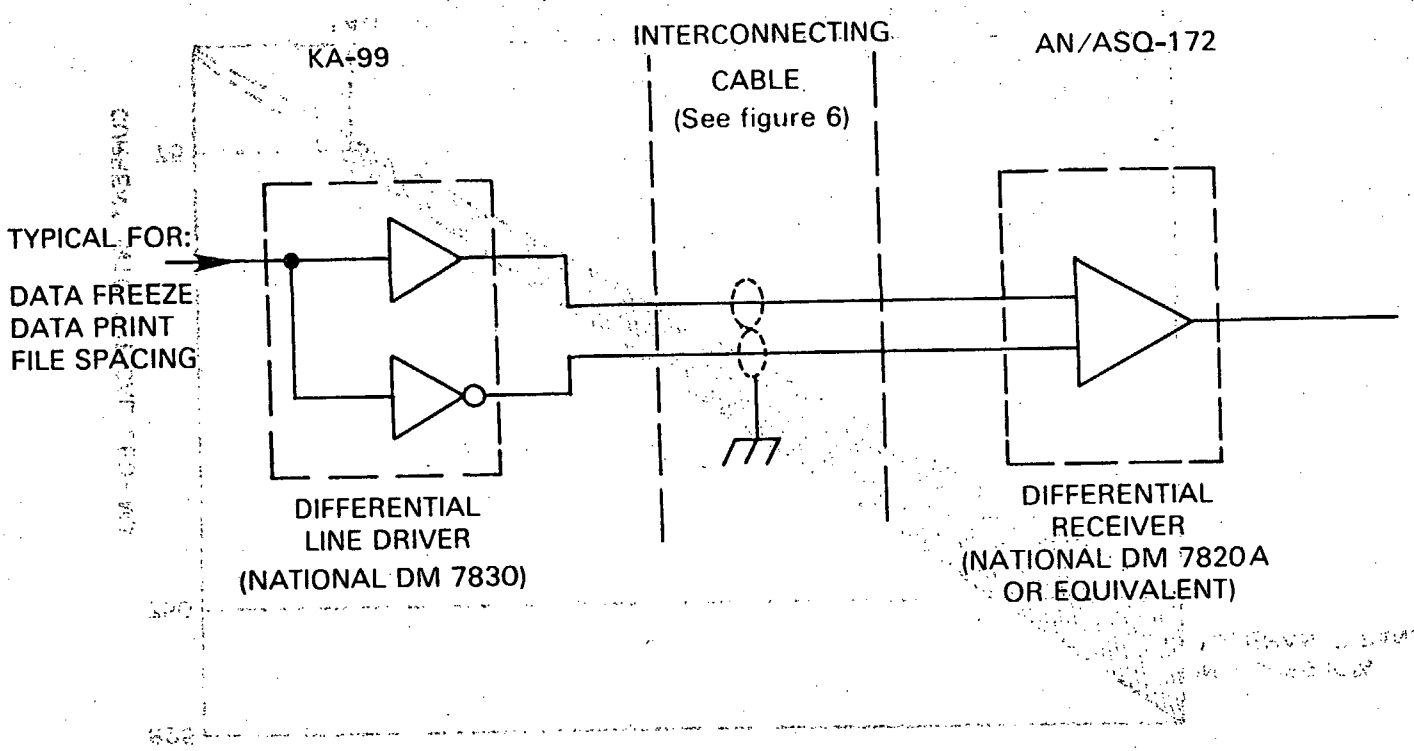


FIGURE 12. KA-99 Output Timing Signal Interface Circuitry.

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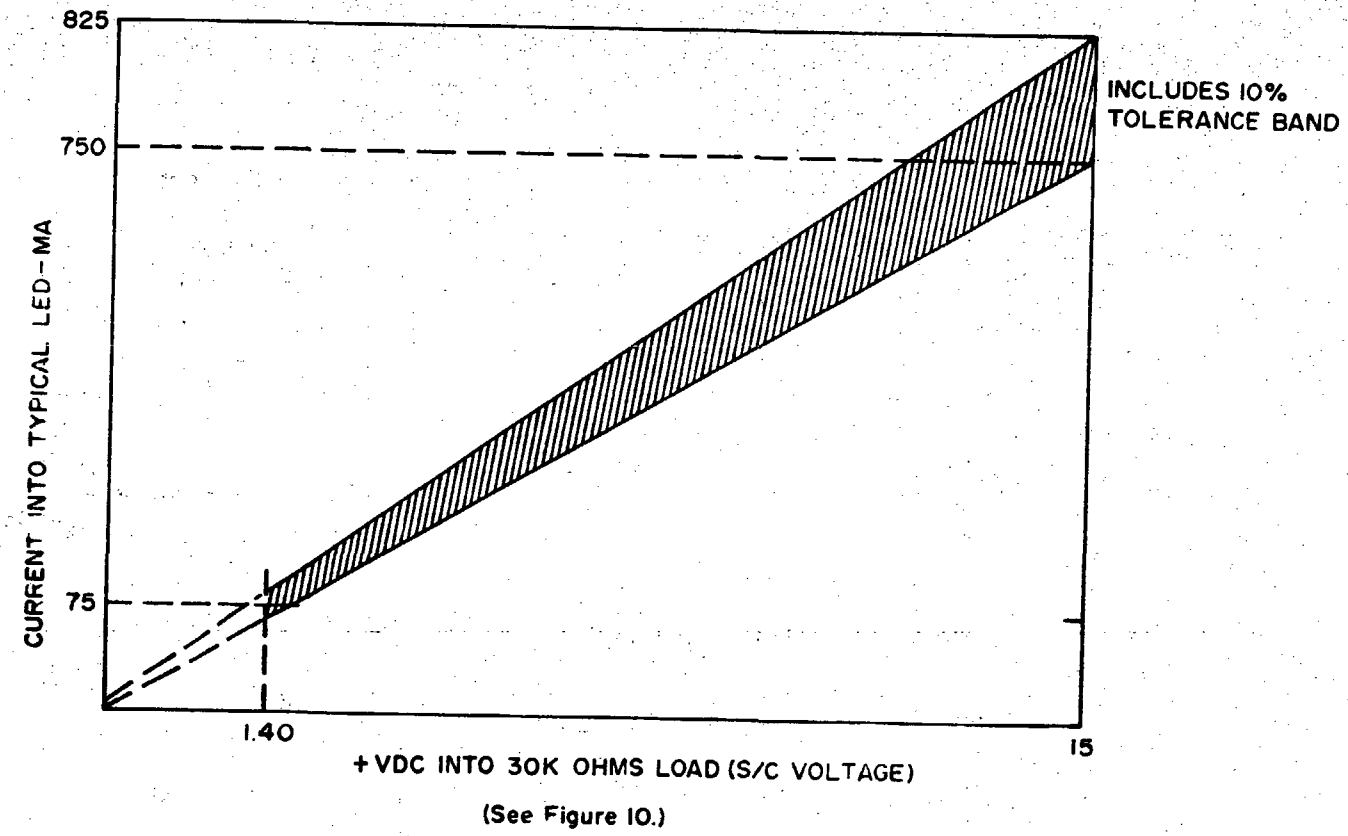


FIGURE 13. Film Sensitivity Control.

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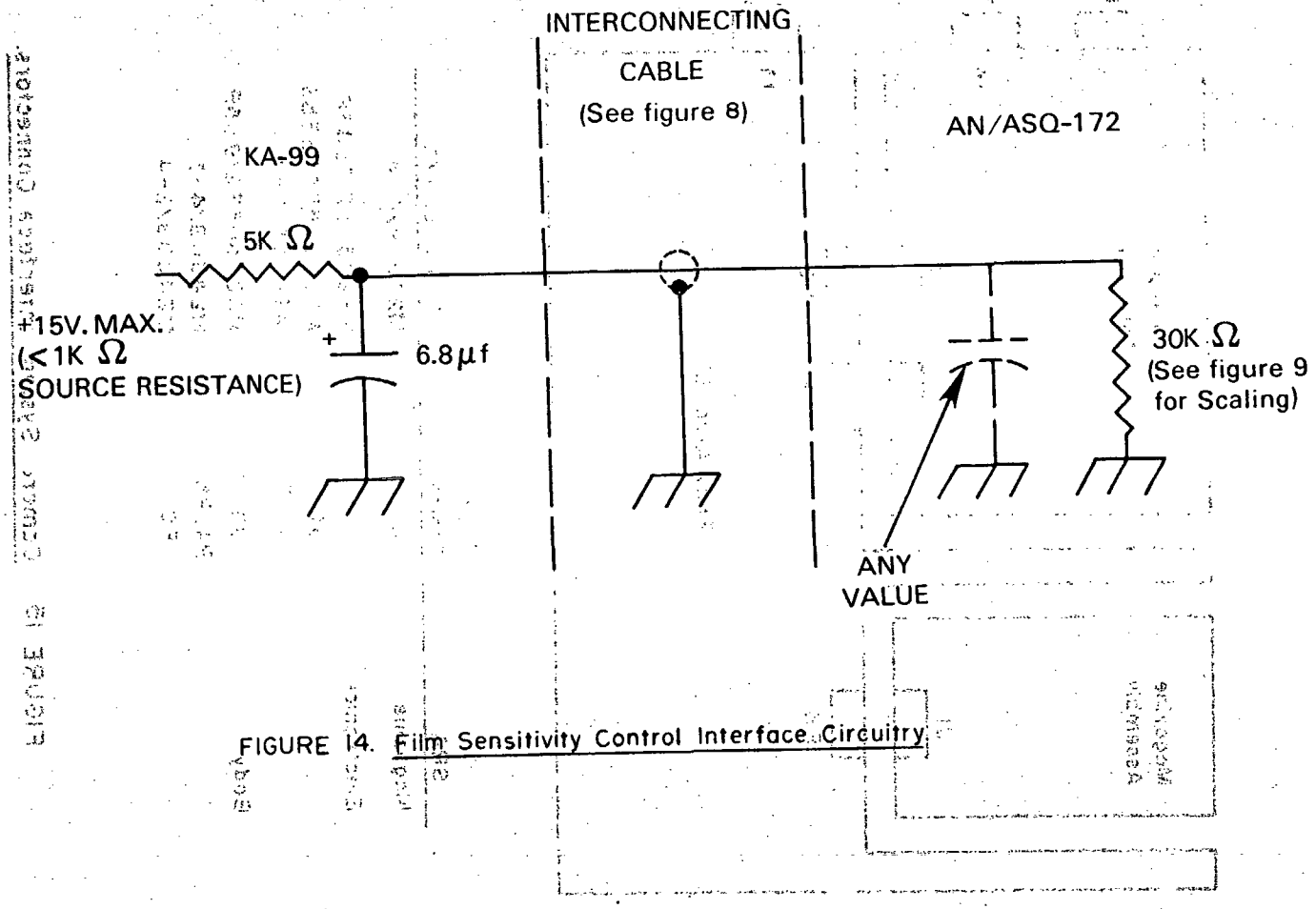
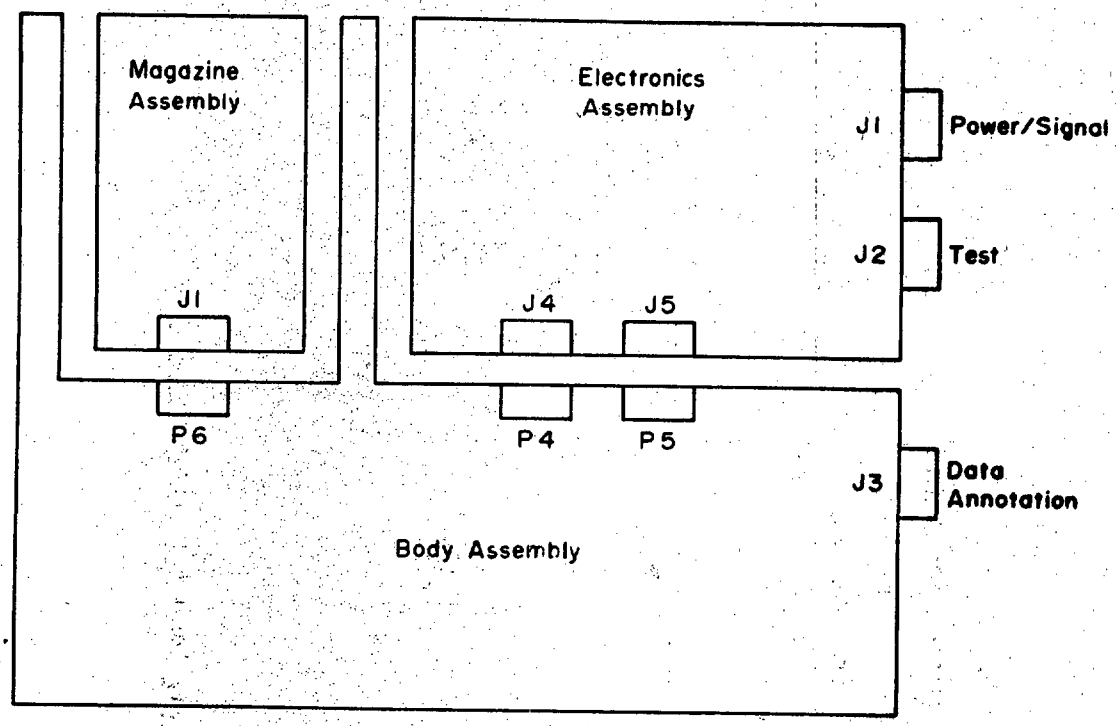


FIGURE 14. Film Sensitivity Control Interface Circuitry.

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SRA	Ref Design	Connector P/N
Magazine	J1	M24308/4-4
Electronics	J1	MS27505E17B35P
	J2	MS27505E25B35S
	J4, J5	M24308/2-5
Body	J3	MS27505E21B35P
	P4, P5	M24308/4-5
	P6	M24308/2-4

FIGURE 15. Camera System Interface Connectors.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: MIL-C-85248(AS) Camera, Still Picture, PANORAMIC, KA-99 ( )

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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